

PFPA Training Course

TO PREPARE FOR WORKING IN FINANCIAL INDUSTRY

An In-depth Introduction Best Real World Practices

Market-Based Credit Risk Rating Model for Public Companies Course #2



PFPA
PACIFIC FINANCIAL PROFESSIONALS ASSOCIATION

Disclaimer

All views expressed in this presentation belong to the author, and do not necessarily reflect the opinions of author's employer or organizations

Contents

- Introduction to Credit Risk Management, as well as LLM/AI/Function calling(context engineering)/MCP/python fastMCP/MCP SDK
- Introduction to Financial Market
- Data and Automation
- KMV/TiC/Agency/Bank Credit Rating
- AI Implementation Automation
- Rating Conversion: PIT to TTC

Introduction to Credit Risk Management and LLM

Credit Risk In Commercial Banks

- It is the largest risk
 - What is a Credit?
 - ✓ Credit is about lending, in particular, loans
 - Term loans, Line of Credit, Commitment, Standby Letter of Credit
 - Often 80-90% of bank's asset
- How the credit risk is managed?
 - Two main requirements and regulations
 - ✓ Capital Management
 - ✓ Loss Reserve (Allowance) Management
 - Involves compliance, policy and procedure, accounting, finance, risk management, analytics, data, system and performance measurement

Credit Risk In Commercial Banks

- Credit Risk Management is centered around
 - Basel II: Capital adequacy and estimation
 - ✓ Basel Risk Parameters and credit risk management language
 - **Credit Rating**, PD/LGD/EAD/Correlation
 - They are fundamental building blocks
 - CECL: Loss Reserve calculation
 - Stress Testing
 - ✓ Finance system failure risk
 - Model Risk management
 - ✓ New Risk from last financial crisis

Credit Risk In Commercial Banks

- Credit Risk Management is centered around
 - They all developed around **Credit Rating**
- **Credit Rating**
 - Credit Risk is the risk that the company won't be able to repay its debt obligations
 - Credit Rating is an assessment regarding company's ability to fulfill its debt commitments
 - ✓ It often assess the Probability of Default – the chance that the debt won't be paid in full
 - Who needs credit ratings?
 - ✓ Banks, Lending Companies, Market Analysts, Stock & Bond Pricing, Portfolio Investment, Risk Management, etc.

Credit Risk In Commercial Banks

- Who generates credit ratings?
 - Banks
 - Rating agencies (Moody's, S&P, Fitch)
 - Finance Companies
 - Academic Researchers
 - They may use different data and different approaches so all ratings are not fully comparable.

Credit Risk In Commercial Banks

- 2012: SEC report to US Congress
 - Can all these ratings be standardized/unified?
 - ✓ Inconsistent across institutions
 - ✓ Inconsistent across time by the same institution
 - ✓ Inconsistent in credit view: Point-In-Time (PIT) vs. Through-The-Cycle (TTC)
 - SEC's answer: seems impossible

Credit Risk In Commercial Banks

- My Research
 - WFC 2021
 - NYU 2019
 - Many Academic Institutions
- Types of credit ratings?
 - Rating agencies: Process-specific
 - Banks: Distribution-Specific (historical data-based; TTC)
 - Academic Researchers: Default-Specific (market data-based; PIT)

Credit Risk In Commercial Banks

- Agency Rating Example: Moody's Historical Default Rates for B-rated Companies B

Moodys B	2001	2002	2003	2004	2005	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016
Historical PD	9.8%	5.4%	2.8%	0.7%	1.3%	0.0%	2.1%	7.6%	0.5%	0.1%	0.5%	0.9%	0.3%	2.3%	1.6%

- Credit Rating is the essential piece for determining the Expected Loss of a credit
 - Allowance

Credit Risk In Commercial Banks

- Bank Ratings
 - Historical data based
 - PD vs. LGD
 - ✓ Unsecured vs. Secured
 - Regression-based
 - ✓ Logistics Regression
 - ✓ Probit Regression
 - Commercial vs. Retail
 - ✓ Commercial: Companies
 - Large, Public, Private, Medium, Small
 - No Data??
 - ✓ Retail
 - Mortgage, Credit Card
 - FICO:
 - Experian/Equifax/TransUnion

Credit Risk In Commercial Banks

- Academic Institutions
 - Market data based
 - ✓ Short-term
 - ✓ Public companies
 - ✓ Unsecured
 - Default Theory-based
 - ✓ Asset falls below the debt
 - Very unstable
 - Difficult to translate into bank ratings
- Example: KMV EDF
 - Default Theory:
 - ✓ Equity is a Call Option with Debt being the strike

Large Language Model

What is a LLM?

- Tokens
 - Fundamental units of text (for languages, coding, images, voices, videos, etc.)
- LLM: Seamless communication between human (natural language) and computer (Coding)
 - ✓ Decoder
 - ✓ Word chain game (of tokens) based on certain rules
 - Probabilistic Generation
 - Context Attention Correlation
 - ✓ Encoder
- An Example
 - An Apple is a

What is a LLM?

- How to derive Tokens?
 - Data Training using Neural network algorithm
 - ✓ What is an algorithm
 - ✓ What is a neural network
 - ✓ Use as much data as possible
- How to derive attention rules?
 - Data Training using correlation matrices
 - Length of Context window

What Is Algorithm?

Algorithm

- Example of algorithm

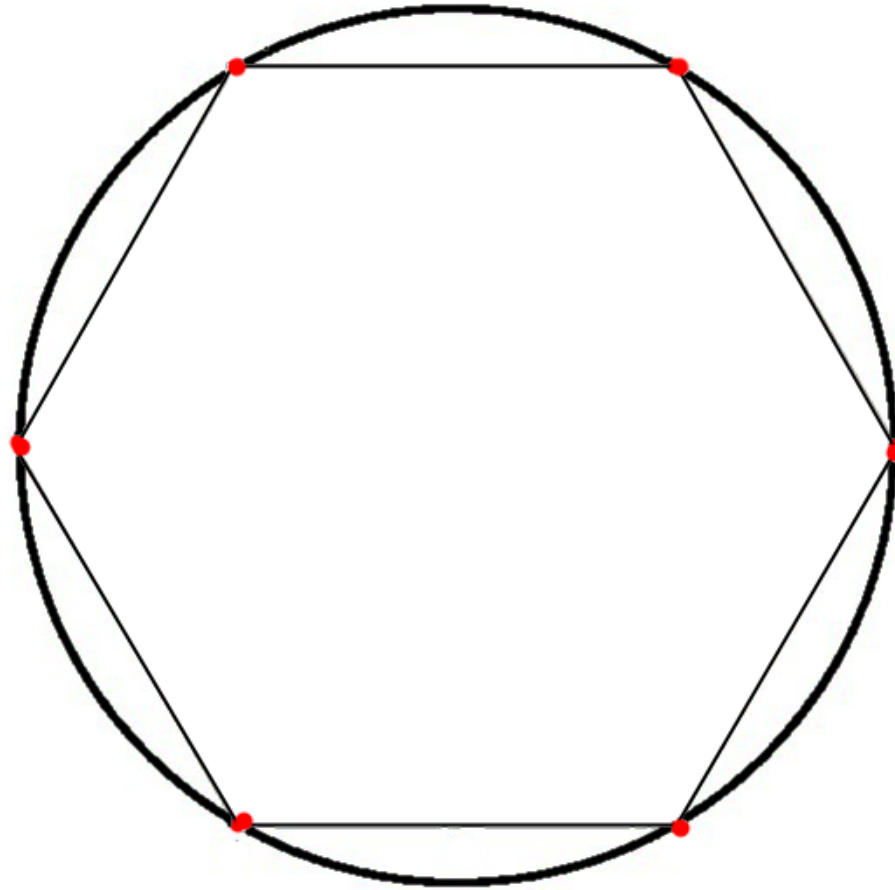
$$\frac{357181}{17} = ?$$



$$\begin{array}{r} 21010 \\ 17 \overline{) 357181} \\ \underline{34} \\ 17 \\ \underline{17} \\ 018 \\ \underline{17} \\ 11 \end{array}$$

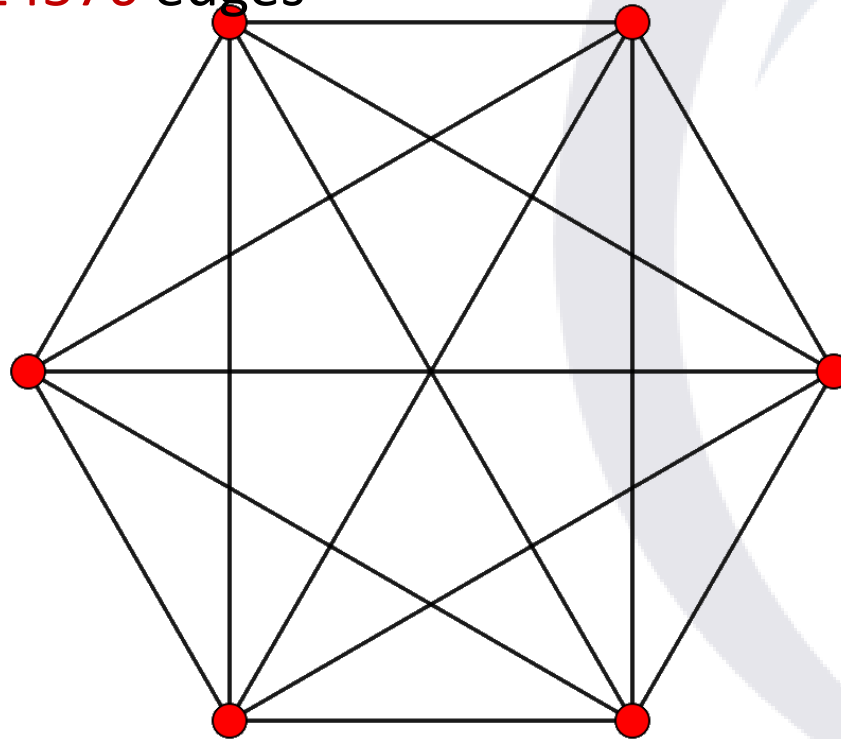
Algorithm

- Another Ex



Algorithm

- Another Example: Calculating π
 - 1500 years ago:
 - Having 24576 edges



Algorithm

- Another Algorithm for Calculating π

1

- What's powerful is the knowledge behind algorithm, not its execution. AI is the **execution of algorithms**.

Origin of Algorithm

- Entscheidungsproblem (decision problem) by David Hilbert in 1928 (Can **machine** always process logics to produce decisions - computability Y/N?)
 - Uncomputability by Turing in 1936
 - Decision process is NOT an algorithm
- Alan Turing Complete System (1936)
 - Universal Turing Machine: all algorithms solvable
 - In other words, algorithm is what executable by a computer
- Rice's Theorem
 - Outcomes of a Turing machine are undecidable to Turing machines
 - Turing machine cannot answer questions about the behavior of other Turing machines

A Mother Algorithm

- Newton's **Root Finding Algorithm** in 1669
 - For example, to solve the equation

$$f(x) = 0.1x^3 - 0.25x - 1 = 0$$

- **Newton Algorithm**

- Take an initial value x_0
- Use the following formula iteratively to find $x_1, x_2, x_3, \dots$

$$x_{next} = x_{prev} - \frac{0.1x_{prev}^3 - 0.25x_{prev} - 1}{0.3x_{prev}^2 - 0.25}$$

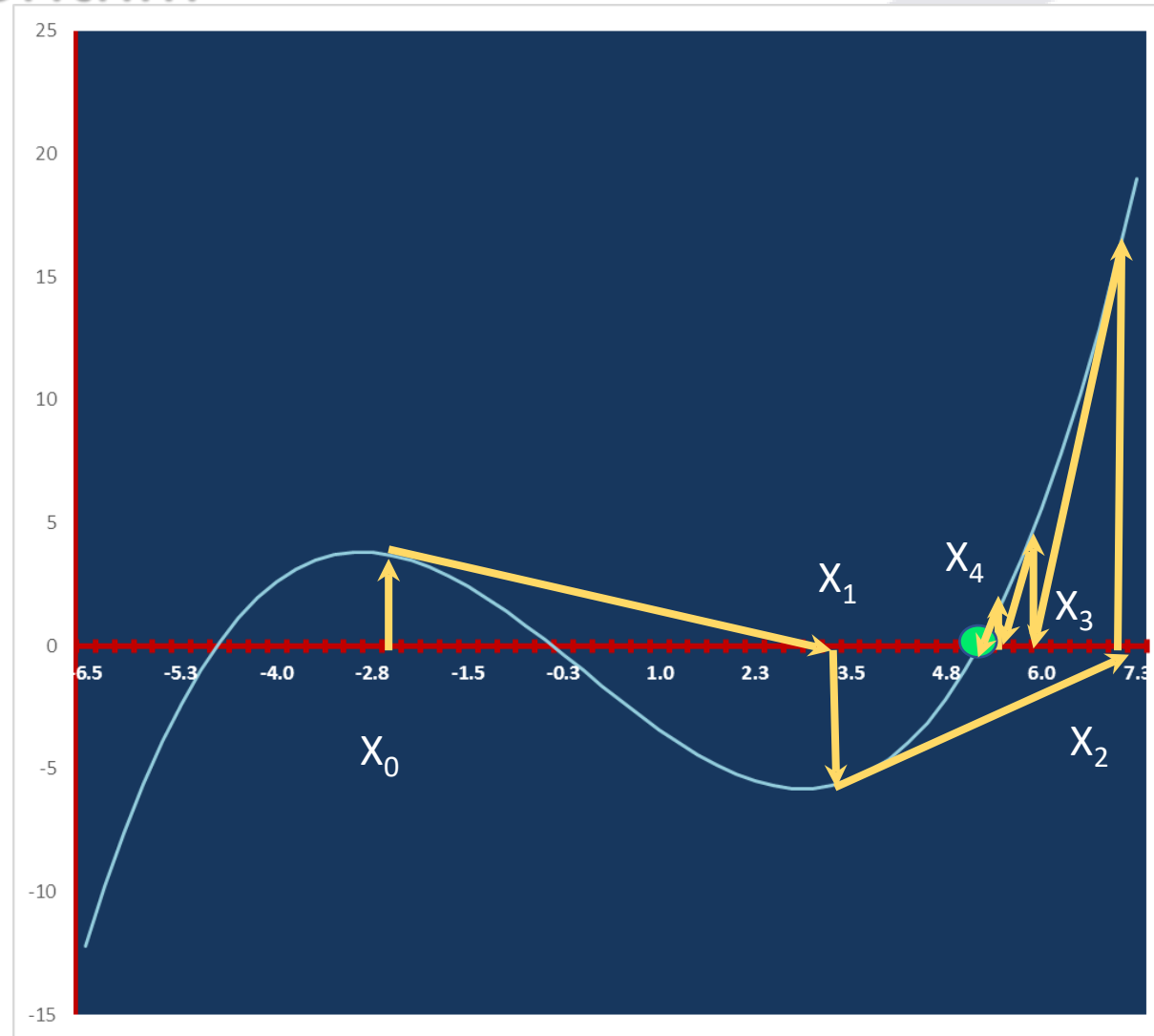
- Let us take an initial value $x_0 = -2.5$

A Mother Algorithm

		$f(x)$
x_0	-2.5	3.7
x_1	3.629	-5.3
x_2	7.277	19.3
x_3	5.832	4.3
x_4	5.280	0.5
x_5	5.191	0.0
x_6	5.189	0.00
x_7	5.189	0.00
x_8	5.189	0.00
x_9	5.189	0.00
x_{10}	5.189	0.00

- So the solution is $x = 5.189$.

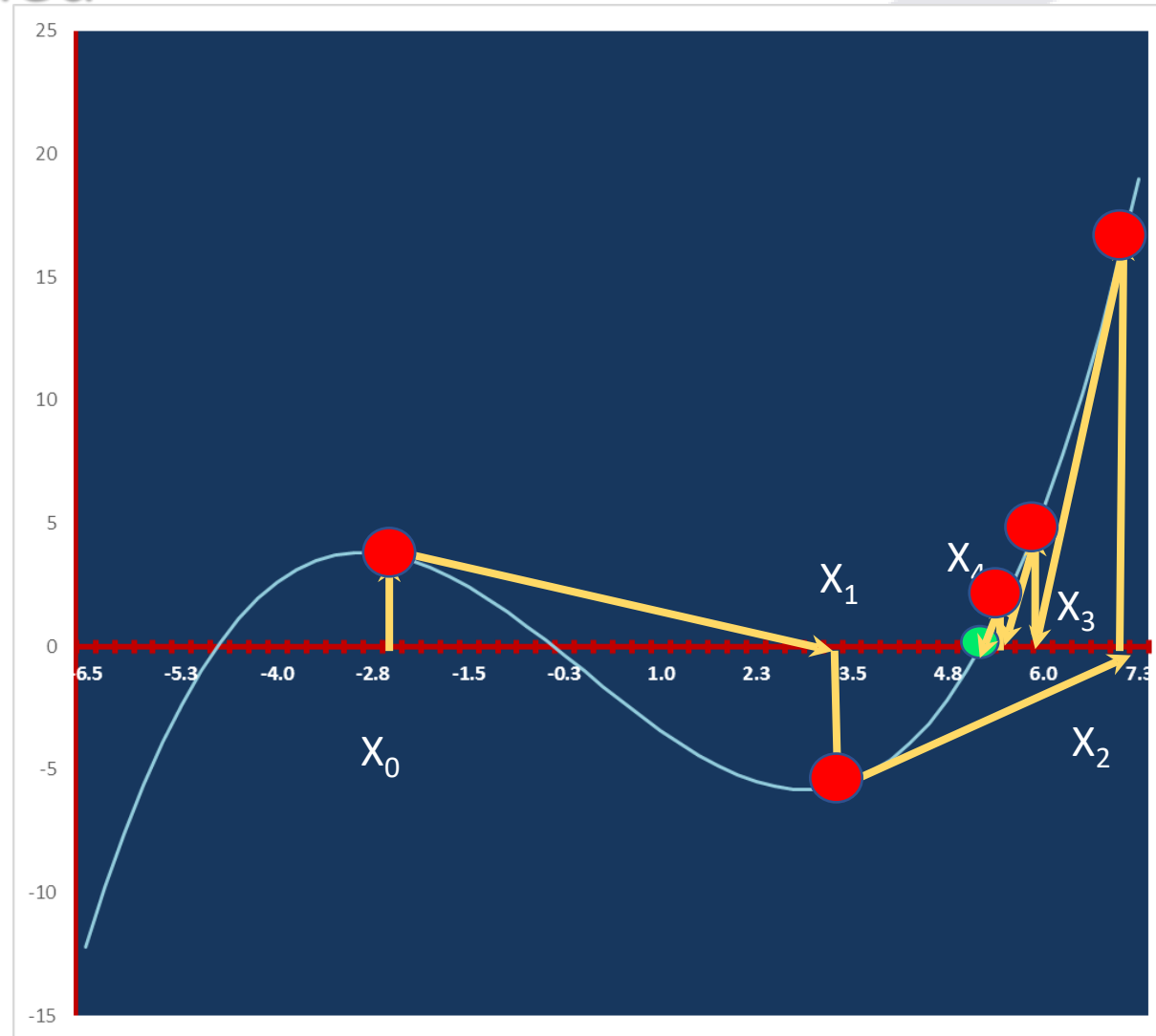
A Mother Algorithm



A Mother Algorithm

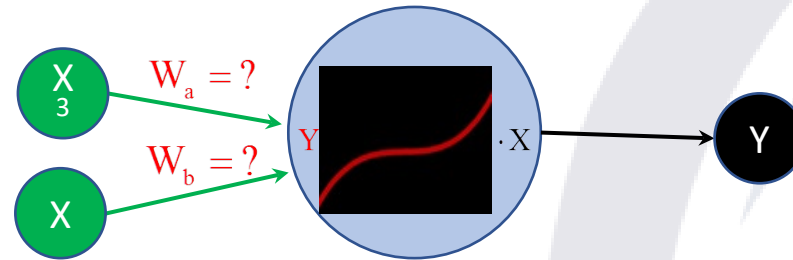
- Newton's Root Finding Algorithm is to walk blindfolded

Walk Blindfolded

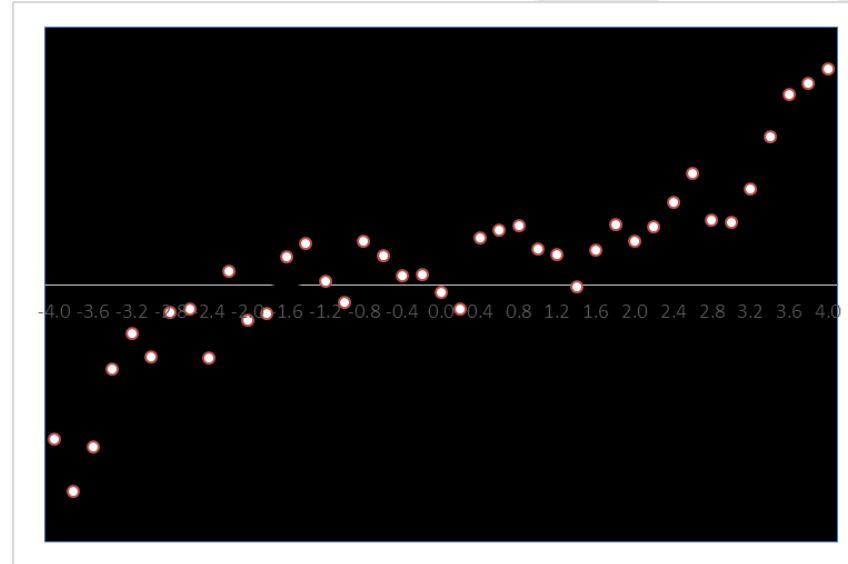


A Training Example

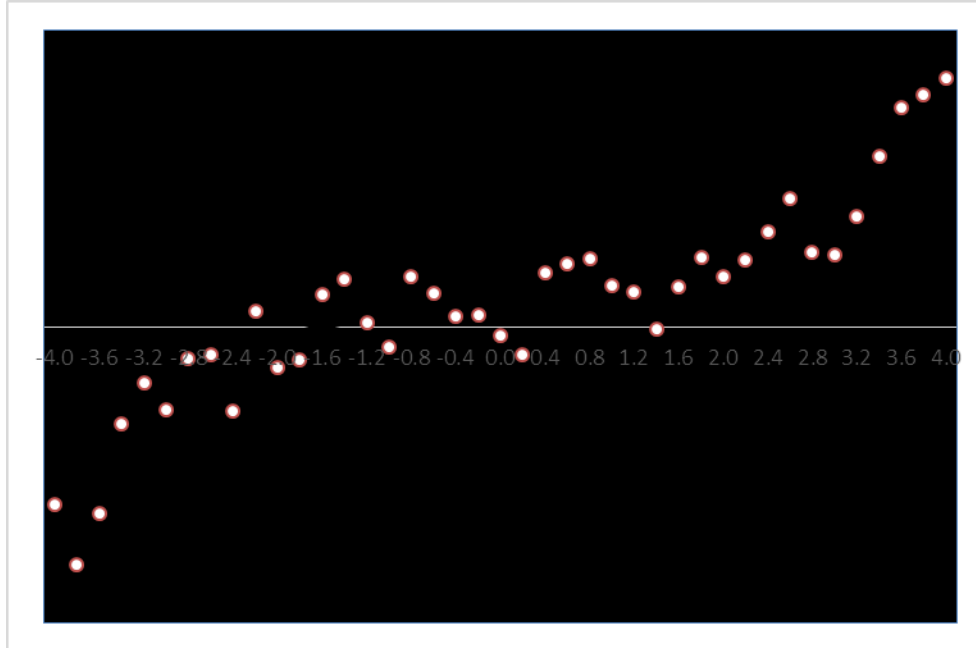
- The neuron



- The Data



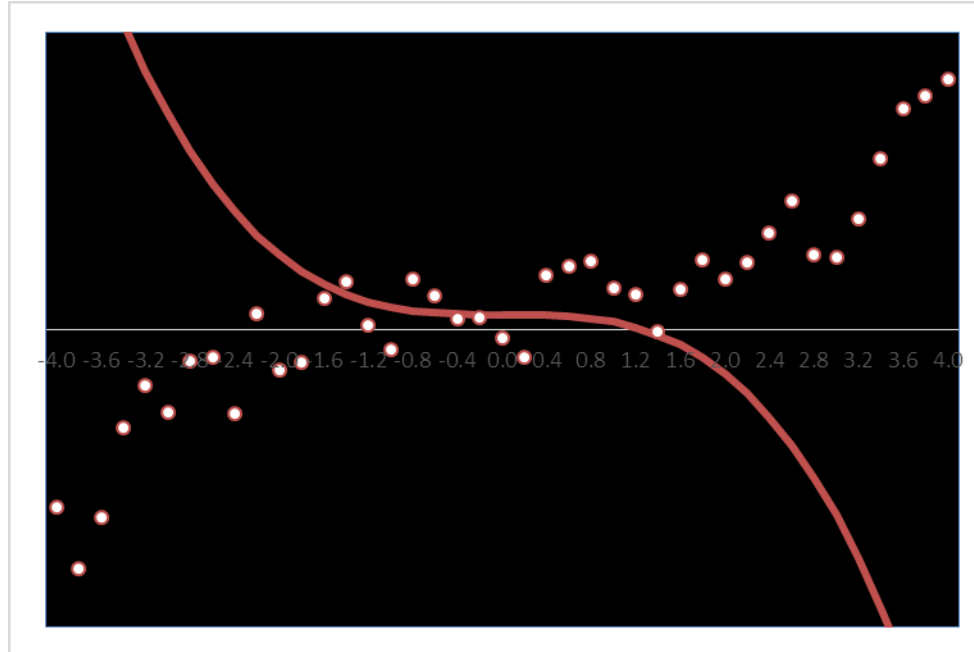
A Training Example



■ Algorithm

1. Start with initial w_0 , initial iteration $k = 0$, calculate initial D_0 , let $S_0=1$ and repeat 2 – 4
2. Set $w_{k+1} = w_k - S_k/2$
3. Calculate D_{k+1} , and $(D_{k+1}-D_k)/(w_{k+1}-w_k)$,
4. Calculate its sign S_{k+1} and set $k = k+1$

A Training Example



Algorithm

1. Start with initial w_0 , initial iteration $k = 0$, calculate initial D_0 , let $S_0=1$ and repeat 2 – 4
2. Set $w_{k+1} = w_k - S_k/2$
3. Calculate D_{k+1} , and $(D_{k+1}-D_k)/(w_{k+1}-w_k)$,
4. Calculate its sign S_{k+1} and set $k = k+1$

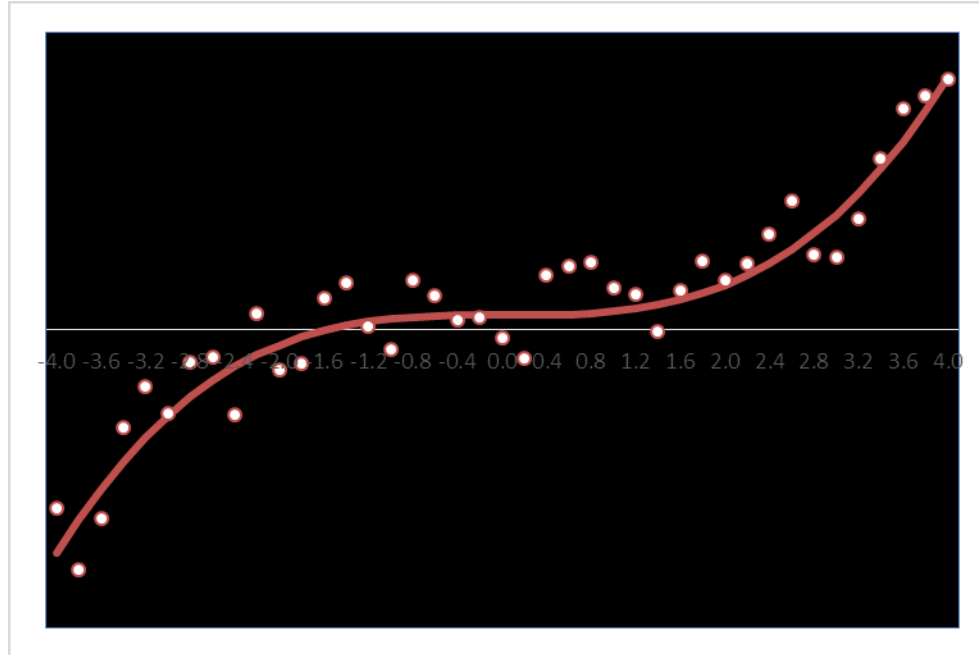
Initial value

$$W_a = -1, W_b = -1$$

Iteration	Wa	Wb	D(Wa,Wb)	Change in W	Change in D	Ratio	Sign
0	-1.0	-1.0	35.05				
1	-1.0	-1.5	35.19	-0.5	0.14	-0.27	-1
2	-1.0	-1.0	35.05	0.5	-0.14	-0.27	-1
3	-1.0	-0.5	34.91	0.5	-0.13	-0.27	-1
4	-1.0	0.0	34.78	0.5	-0.13	-0.27	-1
5	-1.0	0.5	34.67	0.5	-0.11	-0.22	-1
6	-1.0	1.0	34.56	0.5	-0.11	-0.22	-1
7	-1.0	1.5	34.48	0.5	-0.09	-0.17	-1
8	-1.0	2.0	34.42	0.5	-0.06	-0.11	-1
9	-1.0	2.5	34.48	0.5	0.06	0.11	1
10	-1.0	2.0	34.42	-0.5	-0.06	0.11	1
11	-1.0	1.5	34.48	-0.5	0.06	-0.11	-1
12	-1.0	2.0	34.42	0.5	-0.06	-0.11	-1
13	-1.0	2.5	34.48	0.5	0.06	0.11	1
14	-1.0	2.0	34.42	-0.5	-0.06	0.11	1

The Algorithm finds $w_b = 2$

A Training Example



■ Algorithm continues

Iteration	Wa	Wb	D(Wa,Wb)	Change in W	Change in D	Ratio	Sign
0	-1.0	2.0	25.81				
1	-1.5	2.0	34.42	-0.5	8.60	-17.21	-1
2	-1.0	2.0	25.81	0.5	-8.60	-17.21	-1
3	-0.5	2.0	17.21	0.5	-8.60	-17.21	-1
4	0.0	2.0	8.60	0.5	-8.60	-17.21	-1
5	0.5	2.0	0.00	0.5	-8.60	-17.21	-1
6	1.0	2.0	8.60	0.5	8.60	17.21	1
7	0.5	2.0	0.00	-0.5	-8.60	17.21	1
8	0.0	2.0	8.60	-0.5	8.60	-17.21	-1
9	0.5	2.0	0.00	0.5	-8.60	-17.21	-1

■ Algorithm

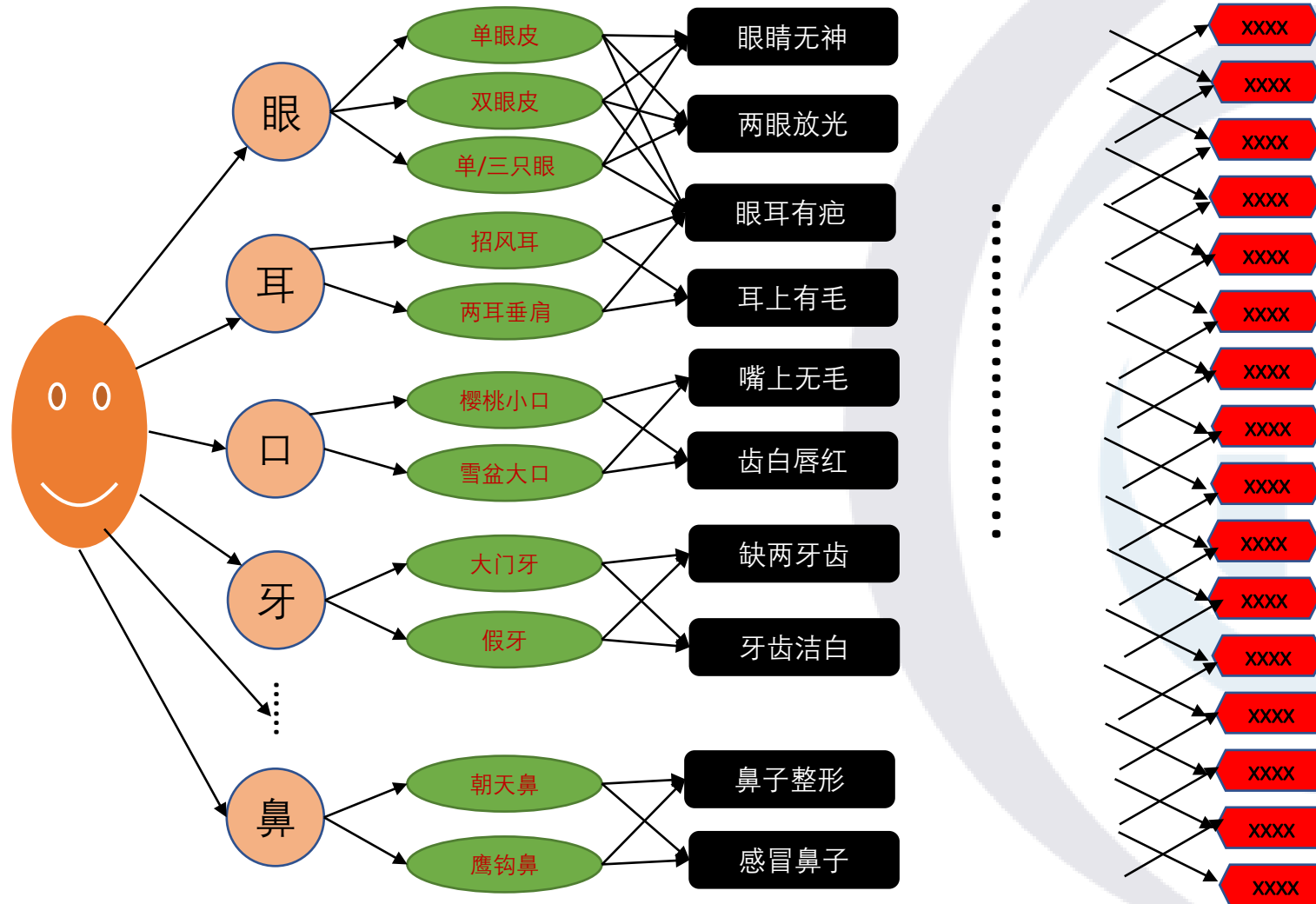
1. Start with initial w_0 , initial iteration $k = 0$, calculate initial D_0 , let $S_0=1$ and repeat 2 – 4
2. Set $w_{k+1} = w_k - S_k/2$
3. Calculate D_{k+1} , and $(D_{k+1}-D_k)/(w_{k+1}-w_k)$,
4. Calculate its sign S_{k+1} and set $k = k+1$

■ It finds $w_a=0.5, w_b=2$

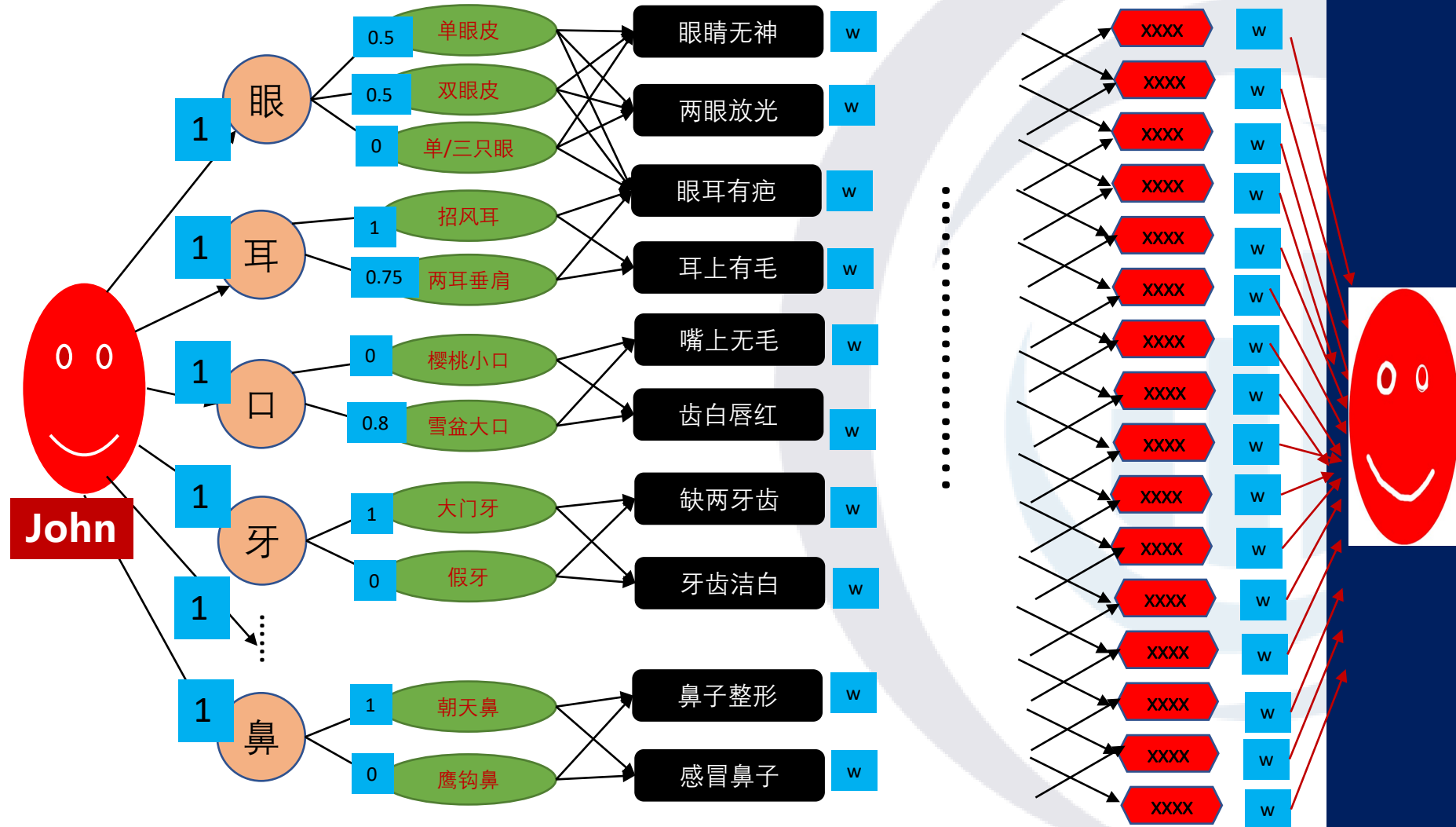
■ The output is $Y = 0.5*X^3+2*X$

What Is Neural Network?

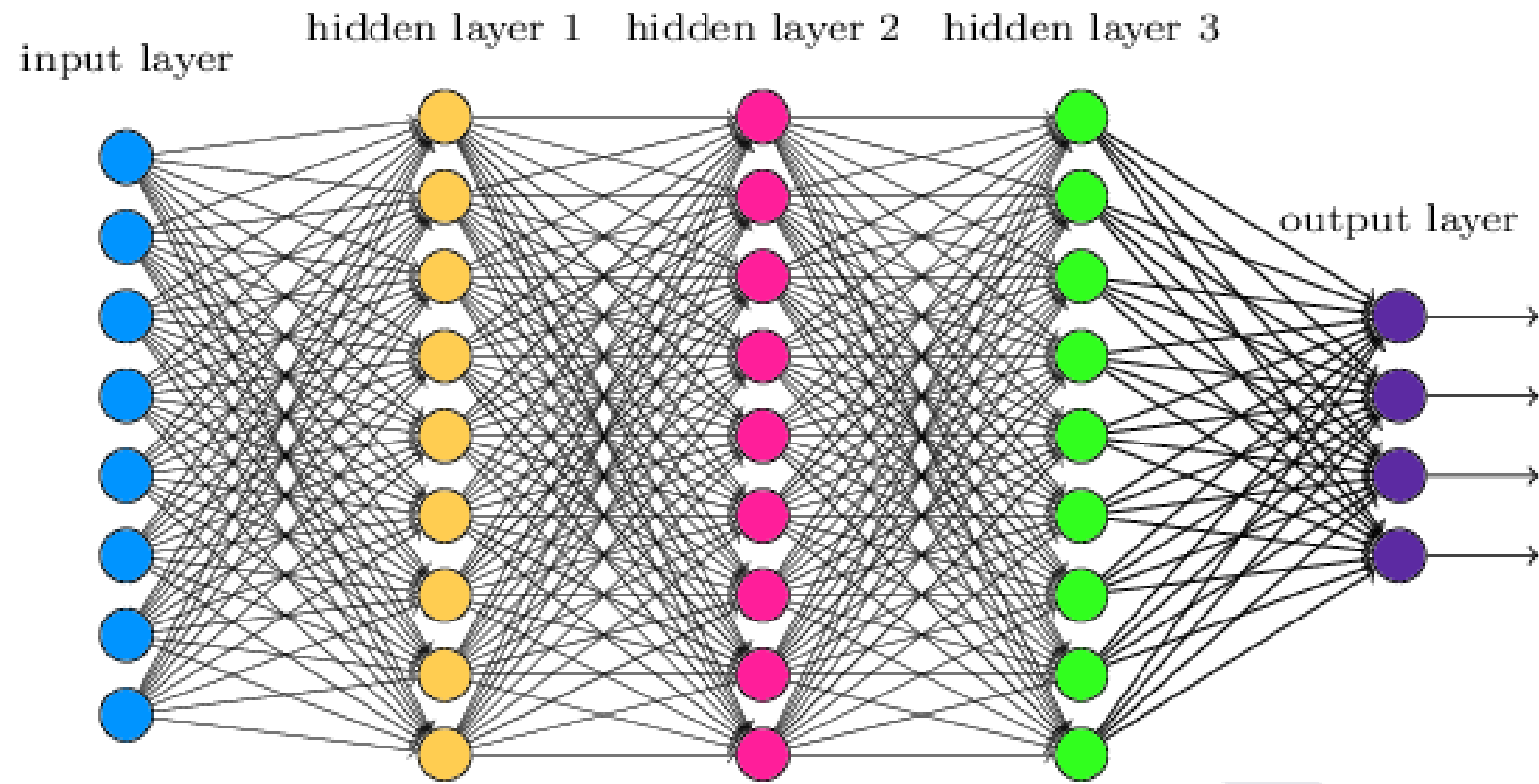
Face Recognition – Training for Tokens



Face Recognition – Weights of a Face after Tokens are derived



Neural Network



Information contained in stock price

Information in stock price

- Stock Market
 - Stock of a company
 - ✓ (also known as equity or share), represents a portion of ownership in the company
 - Stock/share owners are also called investors
 - ✓ For a public company, its shares can be traded (buy/sell) on the stock market
 - Non-public company also has stocks, but difficult to trade
 - ✓ Liquidity & Price are the key differences between public and Non-public companies
 - For public companies
 - Liquidity: tradable anytime
 - Price: observable/discoverable

Information in stock price

✓ Prices frequently change, investors can make profits or suffer losses

○ Stock Market

✓ is the place where public companies list and trade stocks

- Provides liquidity and price discovery
- Requires disclosures of company information
 - Financial information
 - Compliance and regulatory requirements
 - Management information
 - Operational information
 - Businesses and strategies
 - Other information
- Applies and monitors regulations and market disciplines

Information in stock price

○ Why go public?

- ✓ issue stock to raise funds to operate their businesses
- ✓ Exit strategy for early investors
- ✓ To discover market views on the values of the company
- ✓ To access the market to conduct financial and operational activities
 - Issue new stocks or buy back stocks
 - Issue bonds & securities
 - Other financial tools
 - Loans
 - Merger & Acquisition, etc

Information in stock price

- ✓ Types of Stocks

- Common equity: voting right & dividend
- Preferred stocks: no voting rights, dividends with priority

- What moves prices?

- ✓ Any information about the company, the industry, the region, the economy, etc.
- ✓ What does a stock price mean
- ✓ It is the aggregated value of all future cashflows generated by the company
 - The Present Value
 - One of them is the Dividend
 - Some companies don't pay dividends

Information in stock price

- The market value of a company

- ✓ Today's market value is its Total Equity Value (Market Cap)?

- Price is largely driven by Supply and Demand

- ✓ Efficient Market Hypothesis (EMH)

- Rational investors

- Under known information

- Price is efficient: accurately represent the Expected Value of the stock (discounted expected future cash flows)

- Price is “randomly walking” around the expected value

- Today's price is the best forecast for tomorrow's price

- Price reflects all available information

Information in stock price

- ✓ Studied by Eugene Fama, a Ph.D. student at University of Chicago in 1965
 - short-term stock price movements are unpredictable and approximate a random walk
 - Mathematical interpretation will be presented later
- ✓ Eugene Fama won Nobel Prize in Economics in 2013
 - 35 in Economics for Univ. of Chicago, 99 in Total so far



Credit and Debt Market

- Credit/Debt Market

- Bonds and Loans

- ✓ Bonds

- Bond: a financial security under which the issuer promises to pay the holder some amount of money at a later time

- Face Value
 - Interest
 - Future Value

- Government Bond

- Treasury Bonds (Long-term for over 20 years)
 - Treasury Notes (Fixed Rate for 1-10 years)
 - Treasury Bills (short-term for less than 1 year)

Credit and Debt Market

- Credit/Debt Market
 - Municipal Bond
 - Often has tax advantage
 - Mortgage-Backed Bonds (MBS)
 - Pooled mortgages from lenders
 - Corporate Bond: Bond of a company
- ✓ Loans
- ✓ Banks lend moneys to companies or individuals
 - Amount, Interest rate, term, etc.
- ✓ Total Debt: a public company can issue bonds and, at the same time, obtain bank loans
 - Long term debt (more than 1 year)
 - Short-term debt (1 year or less)

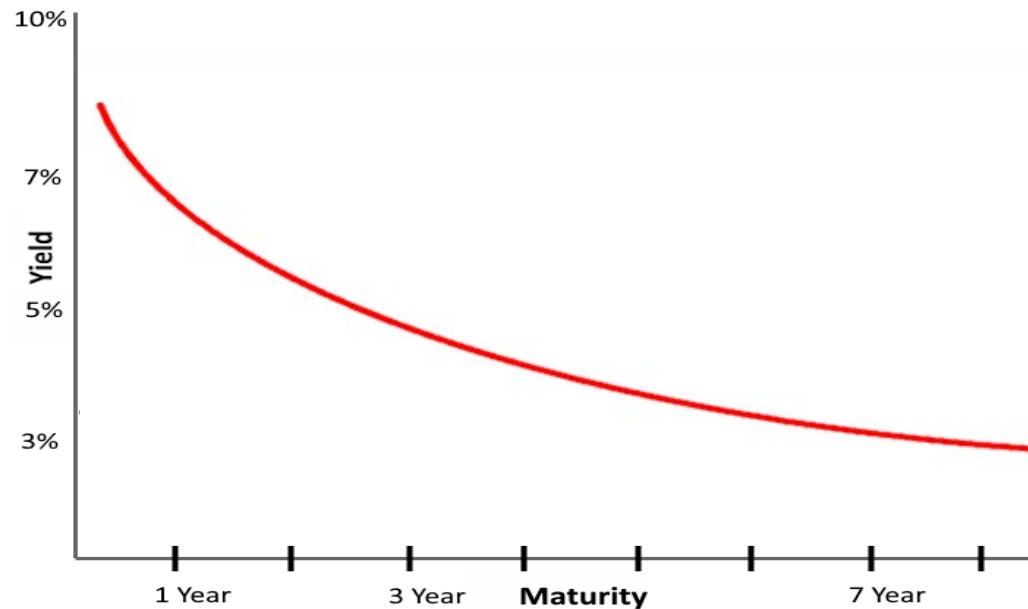
Information in market price

- What if a company does not fulfill its promises to repay the money?
 - This is a risk: the Credit Risk
 - Mitigated through Two methods
 - ✓ Collateral (with Recovery Risk)
 - Many Corporate bonds
 - Mortgages are collateralized
 - ✓ Credit Rating for Creditworthiness (without Collaterals)
 - FICO or other Credit Scores for consumers
 - Credit Card
 - Senior Unsecured Ratings for companies – Default ratings
 - This model is for Default Ratings

Yield curve: relationship between return & term

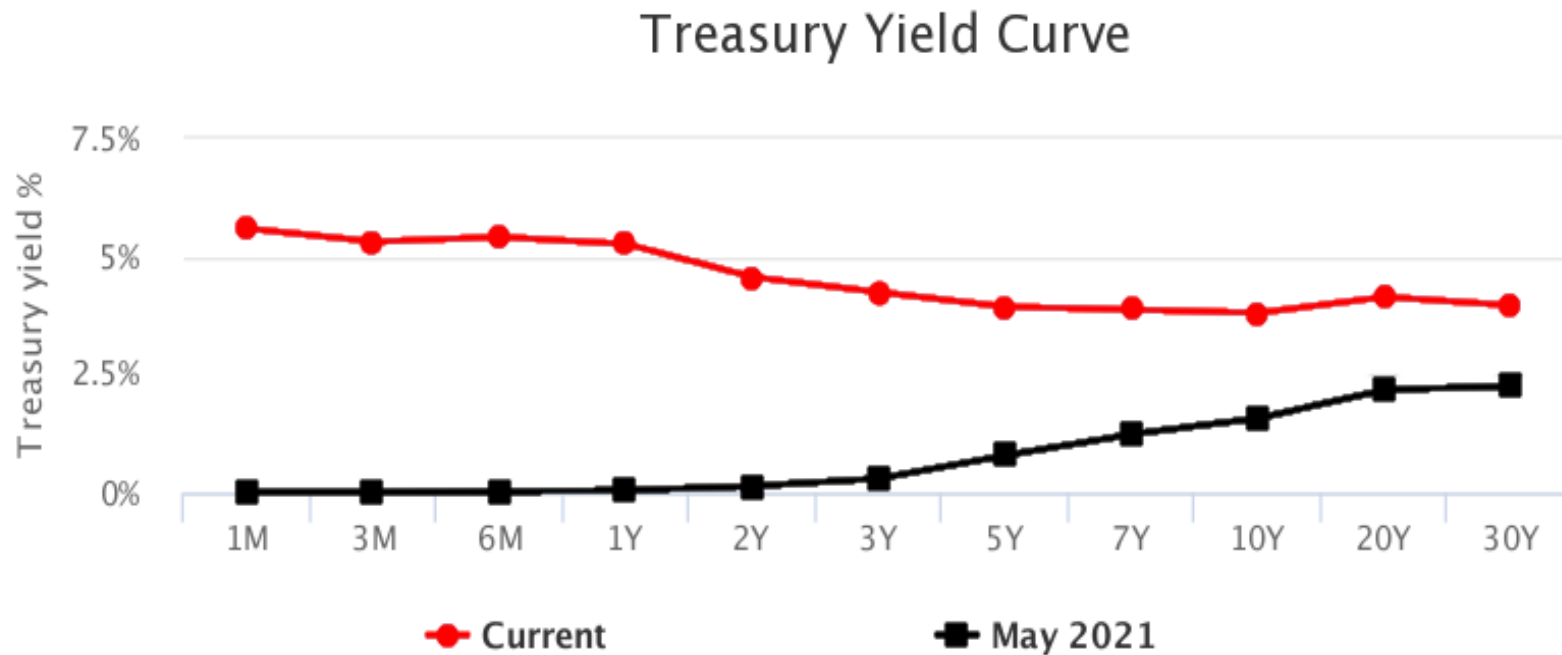
- Yield Curves

- ✓ Relationship between (annualized) returns and maturities of bonds
- ✓ Normal Curve
- ✓ Inverted Curve (possible recession)



Yield curve: relationship between return & term

- Inverted Curve (possible recession)
- 2023/24 vs. 2021. The current curve is Inverted, indicating possible recession



GuruFocus.com

Inverse relationship between interest rate and bond return

- Bond: \$100 face (to be paid in One year)



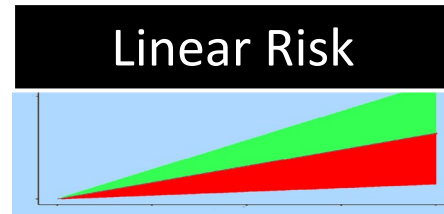
If interest rate = 5%, then Principal = \$95.24, 5% of Interest = \$4.76

If interest rate = 6%, then Principal = \$94.34, 6% of Interest = \$5.66

If interest rate = 4%, then Principal = \$96.15, 4% of Interest = \$3.85

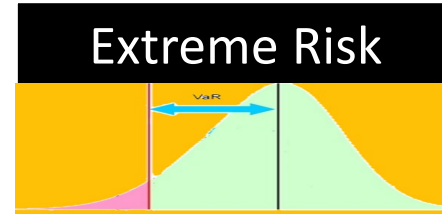
$$\text{Price} = \frac{100}{1 + \text{interest rate}} \quad \searrow, \quad \text{Interest} = \text{Price} \times (\text{interest rate})$$

Three Stages Bank Credit Risk Management



Pre-CRO Time
Concerning Losses

Risks were calculated and managed by individual businesses, then aggregated to the bank level
(Accounting Requirements)



Basel Time
Concerning Insolvency

Risks are calculated by centralized department (Chief Risk Officer) at the bank level
(Capital Requirements)



Stress Test Time
Concerning Financial System Collapses

Risks are forecasted and calculated at both the business and the bank levels for **all bank under the same scenario**

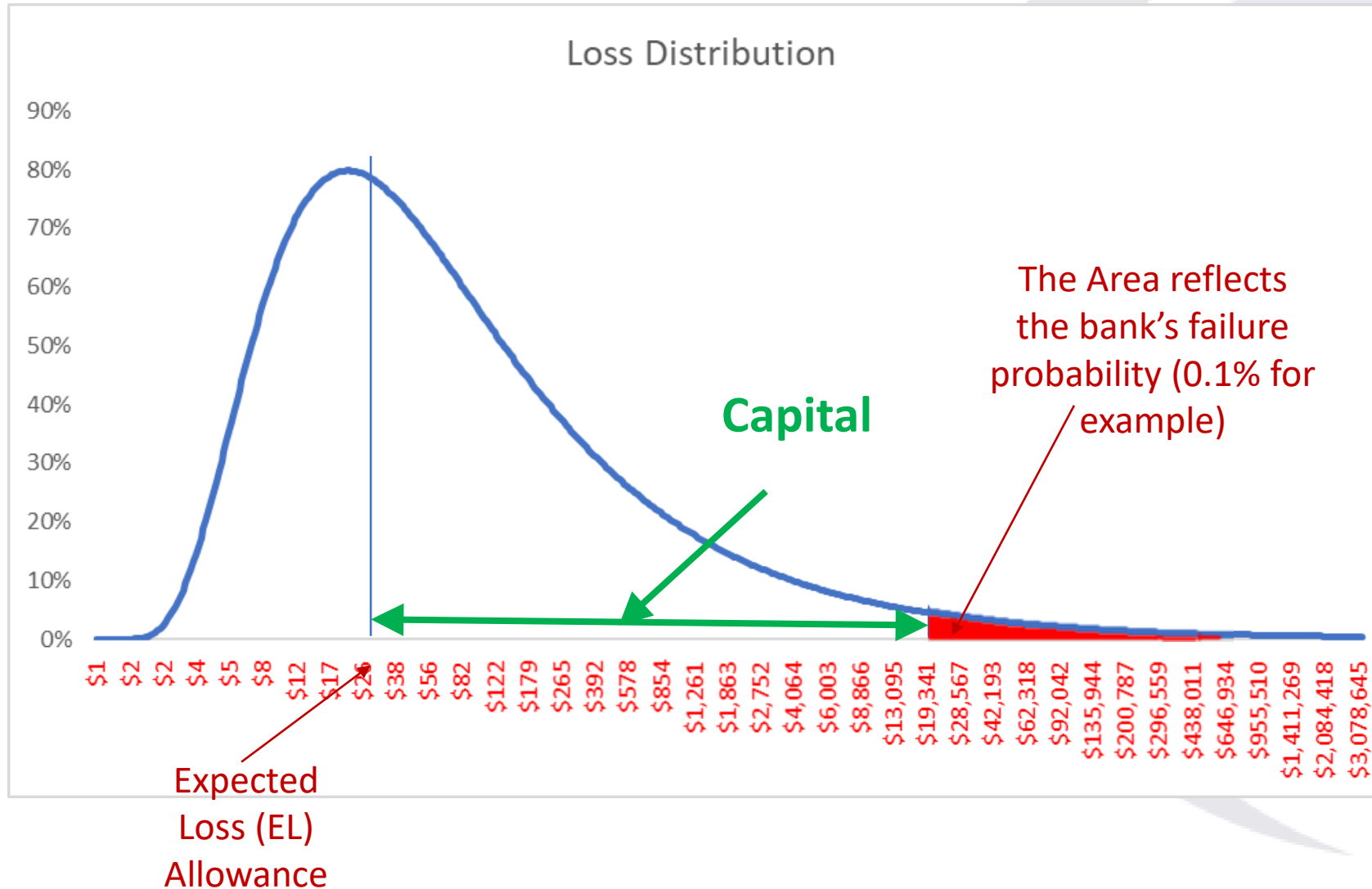
Accounting Rules

1. Additive
2. Cannot book things have not happened, no future items (revenue, etc.)

Loss Allowance

On each loan and aggregated

Loss Distribution: Allowance and Capital



Market value of a company
My China Trip Schedule

Market value of a company

- Price reflects all available information
 - Can we read certain information from price?
 - ✓ For example, credit information?
 - ✓ Before we can do this, we have to understand what is the value of a company because the stock is driven by the company value
 - What value? There are different values
 - ✓ Book Value: company value based on accounting rules
 - Time value of Money?
 - Amortization

Market value of a company

- Market Value: if we sell the company on the market now, how much does it worth ?
 - Liquidity Value/Liquidity Discount
- Future value: company value at a future date. This important when the company has debt (to be paid in the future).
 - Future cashflows are often uncertain, or “risky”
 - Riskiness is measured by “probabilities” associated with the future cashflows
 - The “Expected Value” of future cashflows, which depends on these probabilities, is the Price (or value)
 - However, how to estimate these probabilities?

Market value of a company

- ✓ Accounting often considers no probabilities
 - Cost-based amortization or pre-determined values
- ✓ The Market often uses Risk-Neutral Probabilities (RNP)
 - RNP is based on No-arbitrage pricing (or Arbitrage-free principle)
- ✓ Empirical Probability can also be used. However, the market price may not be consistent with it.
 - It may not contain factors such as market liquidity, interest rates, market return benchmarks, etc.

○

Market value of a company

- Fundamental Accounting Equation
 - $\text{Asset} = \text{Liabilities} + \text{Equity}$
 - ✓ However, this Asset is NOT market value of asset, but accounting asset
 - ✓ This Equity is owner's equity, NOT the market value of the stocks
 - ✓ Time value of money is often ignored
- A Public Company has Market Equity (Stock) and Market Value of Asset
 - $\text{Equity} = \text{Asset} - \text{Debt}$
 - ✓ The Problem: Asset is the market value and is not observable, Debt is to be paid in the future, the Equity is observed today

Balance Sheet of a company

- Balance Sheet for Accounting Equation

- Asset = Liabilities + Equity

- ✓ A liability is what the company owes to others. It is settled over some period of time.

- Loans

- Bonds

- Mortgages: Loans collateralized by real estate

- Accounts payable

- Deferred revenue: Money received before earned it.

- Warranties: money companies owe to consumers for product repairs and replacements.

- Accrued expenses: Liabilities that have built up over time and are due

Balance Sheet of a company

- Balance Sheet for Accounting Equation

- ✓ Types of Liability:

- Current Liability (to be paid within the calendar year)

- Income taxes payable
 - Notes payable or loans from the bank
 - Accounts payable

- Long-term Liability

- Deferred credit
 - Post-employment benefits
 - Warranty liability

- Expense, if not paid immediately, would be part of Liability

Balance Sheet of a company

- Balance Sheet for Accounting Equation

- Expense examples

- Rent
 - Building maintenance and repairs
 - Training and education
 - Utilities
 - Salaries
 - Business and health insurance
 - Software subscriptions
 - Equipment and supplies
 - Marketing and advertising
 - Transportation

- ✓ Debt is borrowed money from another party to be paid in a predetermined term in the future

- is part of Liability:

Market Value of a company's Asset

➤ Short-Term Debt

➤ Long-Term Debt

- **Company's Market Value of Asset**

- $\text{Equity} = \text{Asset Market Value} - \text{Debt}$

- ✓ Shareholders own the residual value of a company after all Debt is paid

- ✓ Debt is to be paid in the future, the Equity is observed today so the calculation is not straight to find the asset

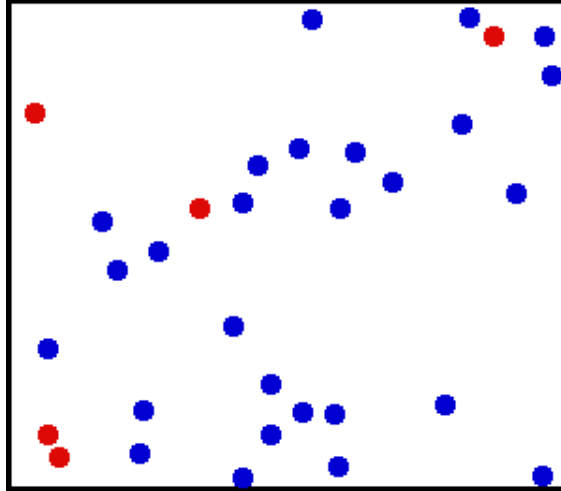
- How to Find the Market Value of Asset?

- ✓ Nobel Laureate Robert Merton: Equity is a Call option on the Asset with Debt being the Strike

Option pricing theory

Option pricing theory

- Stock is a Random Walk
 - Brownian Motion
 - ✓ 1827 Botanist Robert Brown observed
 - Pollen grains suspended in water jiggled about under the lens of the microscope, following a zigzag path

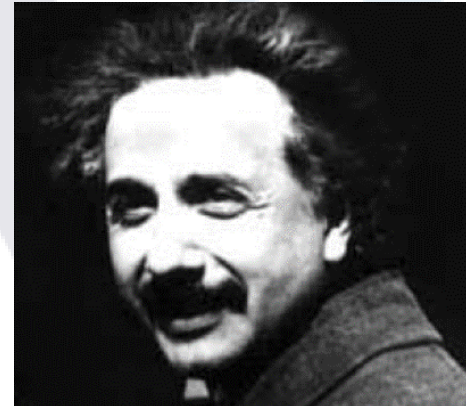


- Even for *Clarkia pulchella* pollen grains that had been stored for a century

Option pricing theory

- Stock is a Random Walk
 - Brownian Motion
 - ✓ In 1905, Einstein discovered for Brownian Motion, theoretical diffusion equation of dynamical equilibrium
 - The displacement of a Brownian particle is proportional to the square root of time

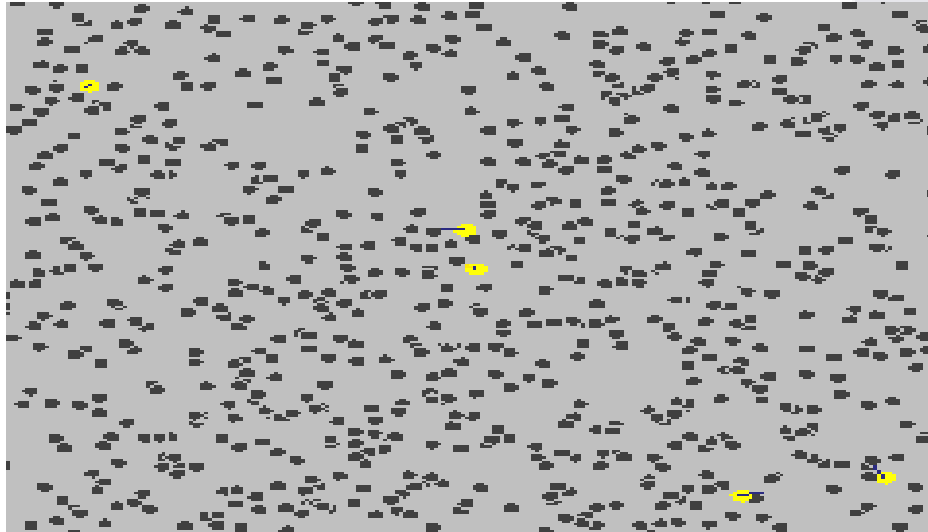
$$\rho(x, t) = \frac{N}{\sqrt{4\pi Dt}} e^{-\frac{x^2}{4Dt}}$$



- Gaussian Normal Distribution!!!

Option pricing theory

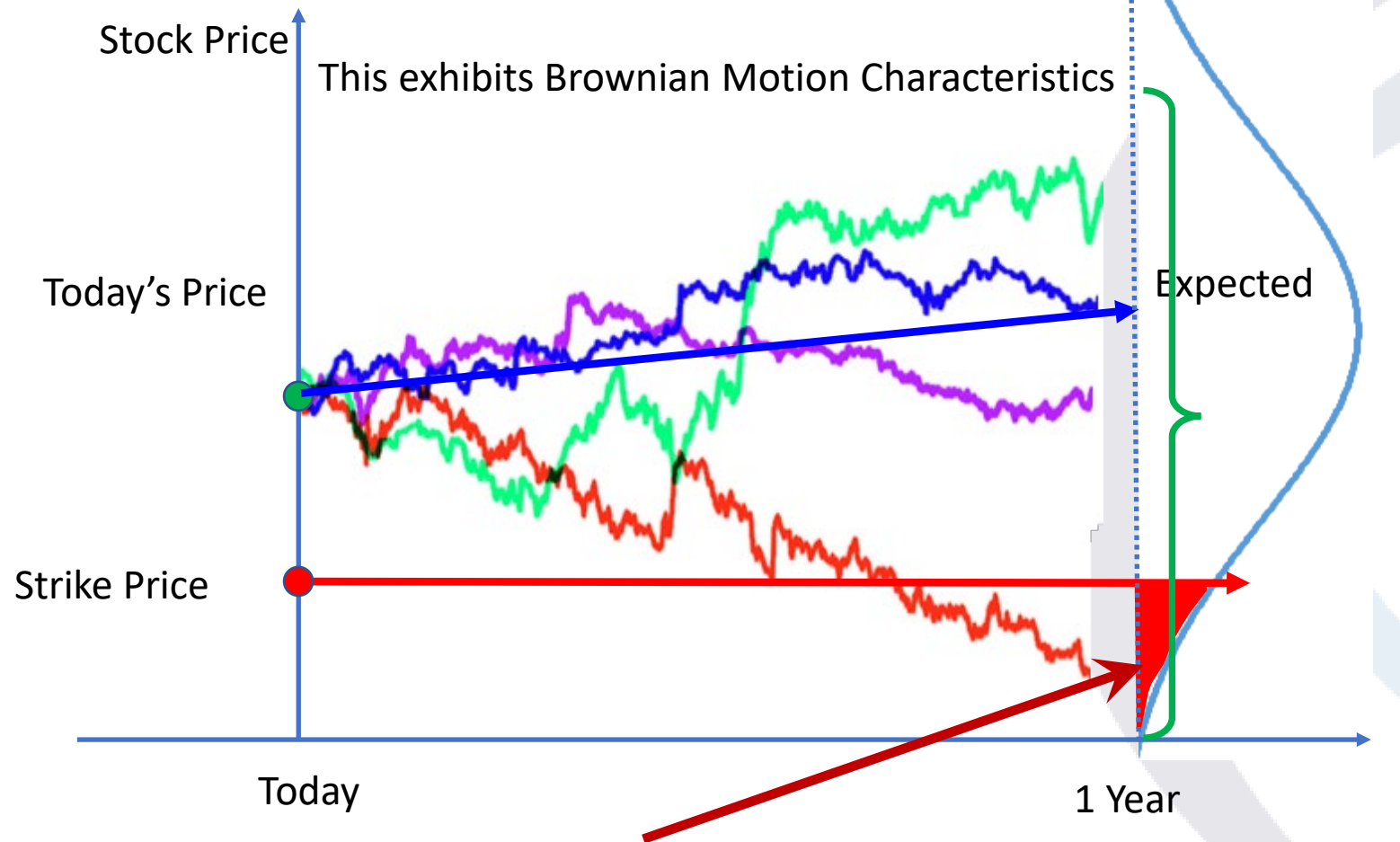
- Stock is a Random Walk
 - Brownian Motion Demystified
 - ✓ Pollen grain's size is about 1/1000 meter in diameter
 - ✓ Water Molecule's size is about 10000 times smaller
 - ✓ Temperature leads to molecular motion
 - ✓ Collision at the rate of millions/second



Option pricing theory

- **Brownian Motion & Mathematics**
 - N. Wiener in 1923 developed Wiener Process
 - ✓ Existence and construction of Brownian motion was established
 - Ito's lemma in 1944
 - Richard Feynman in 1948 used Wiener integral to develop path integral formulation of quantum mechanics)
- **Brownian Motion & Stock Price**
 - Price = Pollen
 - Individual Investors = Molecules
 - Trades by Individual Investors determine the price: Molecules collide Pollens
 - Therefore Random Walk is applicable to stocks

Option pricing theory



This area represents the chance (or the probability) that the stock falls below the Strike price.

Option pricing theory

- In 1973 Fisher Black, Myron Scholes and Robert Merton developed the equation for hedging the option, and derived, based on Ito's lemma, the celebrated Black-Scholes-Merton formula for Call price

$$P \cdot \Phi \left(\frac{\ln \frac{P}{K} + \left(r + \frac{\sigma^2}{2} \right) T}{\sigma \sqrt{T}} \right) - K \cdot e^{-rT} \Phi \left(\frac{\ln \frac{P}{K} + \left(r - \frac{\sigma^2}{2} \right) T}{\sigma \sqrt{T}} \right)$$

- They received Nobel Prize in 1997



Credit rating theories

Credit Rating theory

- How do we know if the Asset will fall below the Debt?
 - We need to know the Asset's future value
 - We need to know the probability of the future Asset following below the Debt
 - We need to know how the Asset will evolve in the future
 - ✓ Brownian Motion
 - But the Asset is not observable, only the Equity is observed.
 - ✓ Merton says that there is a Call Option relationship exists between the Asset and the Equity

Credit Rating theory

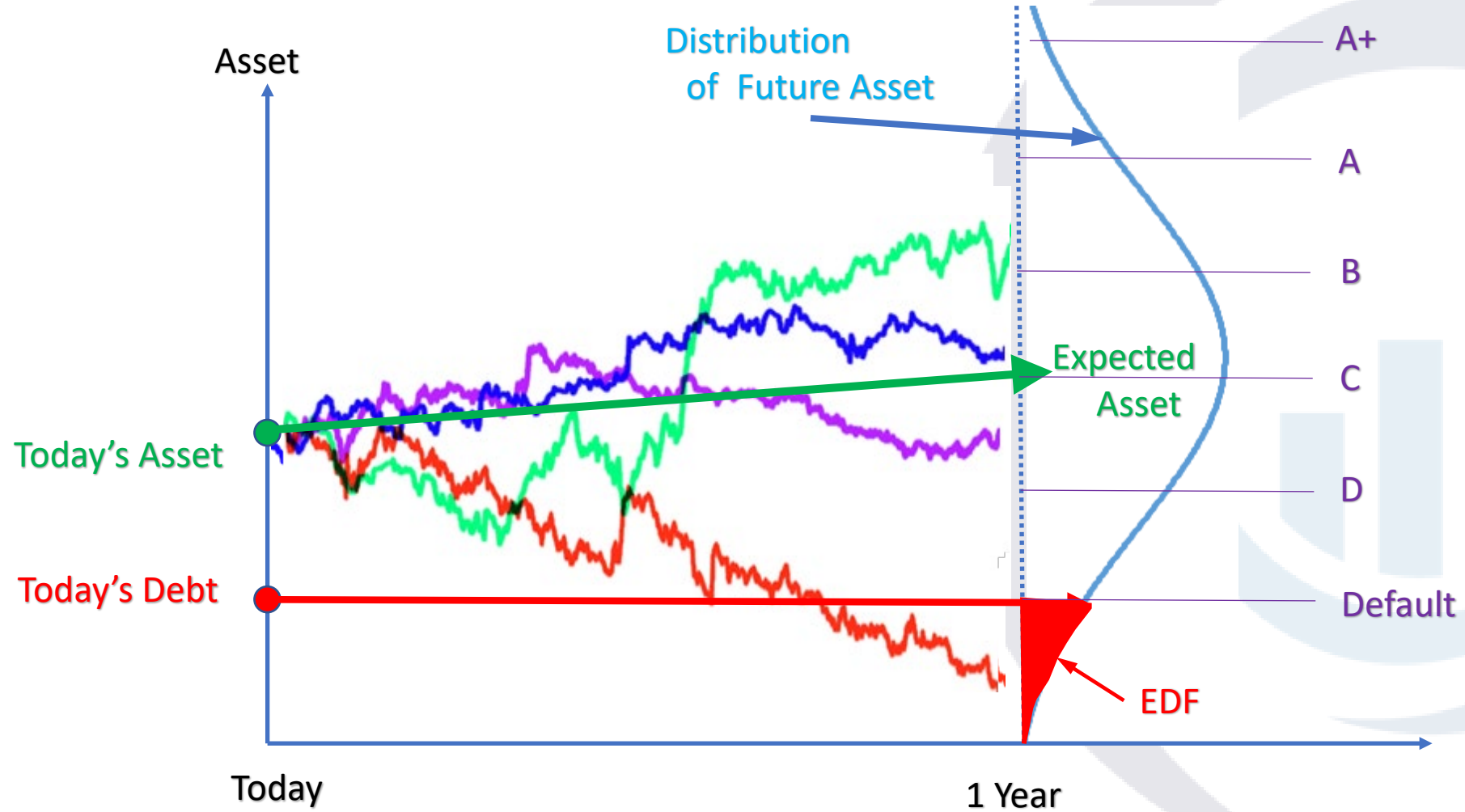
- But the Asset is not observable, only the Equity is observed.
 - ✓ Merton says that there is a Call Option relationship exists between the Asset and the Equity
 - ✓ This relationship can be described using Black-Scholes-Merton Formula
 - ✓ This relationship can be used to solve the Asset from the Equity
- How do we assess credit from the Asset?
 - ✓ We assess the probability of the Asset falling below the Debt using Brownian motion calculus (Stochastic calculus)
 - ✓ Based on this Probability, we derive the credit rating

KMV and TIC rating methods

Credit Rating theory

- This Credit Rating is known as Distance-to-Default (DD) or Expected-Default-Frequency (EDF) rating first introduced by KMV.
 - ✓ The rating is very useful. It is a Point-In-Time (PIT) forward looking rating that presents market's current view on the company
 - ✓ Moody's spent \$ 220 millions to purchase KMV
 - ✓ However, it has known problems
 - Most significant problem: unstable rating with huge volatility
 - It also changes too frequently as it is directly driven by stock prices
 - Not suitable for banks to use as banks often require Through-The-Cycle (TTC) ratings

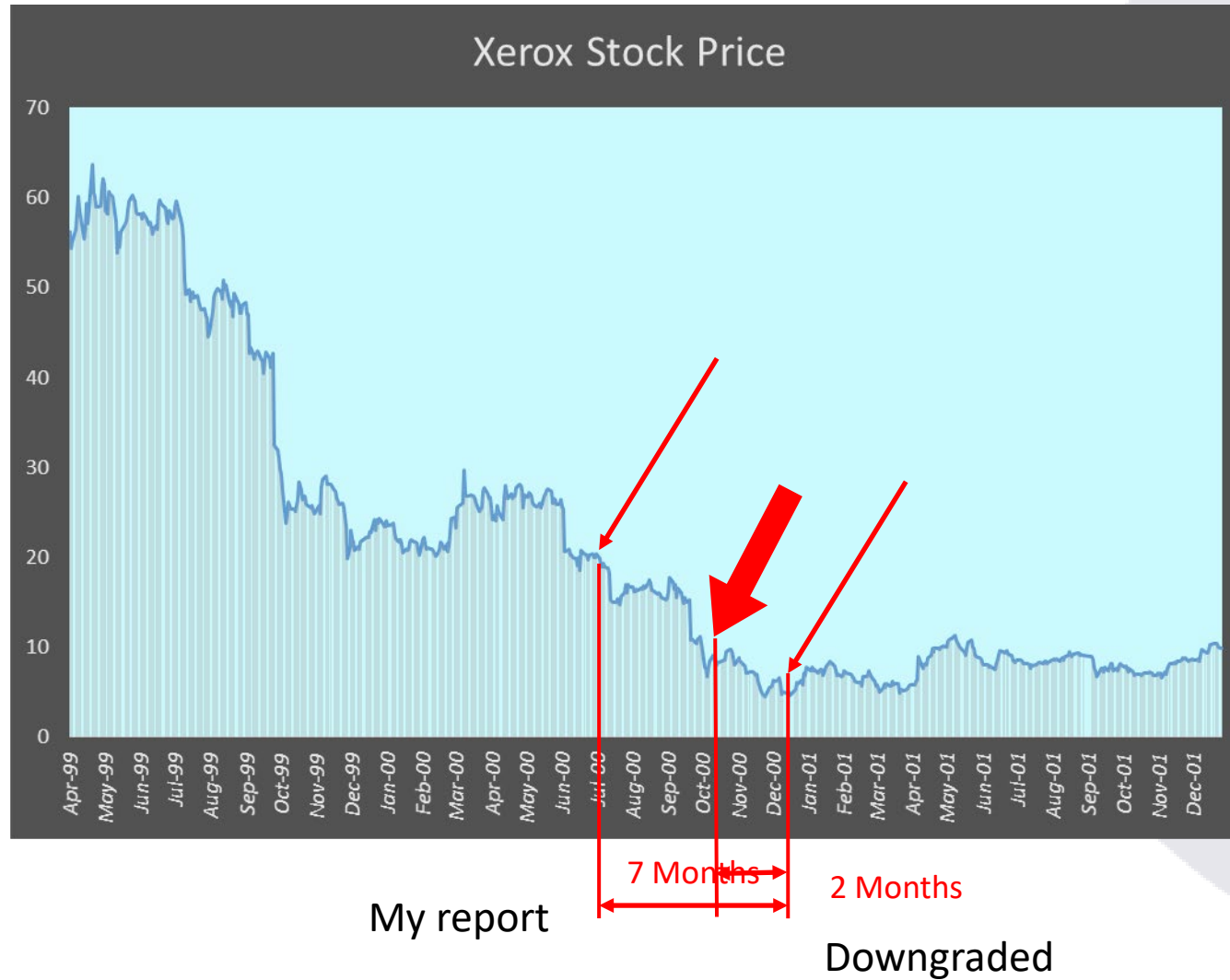
Credit Rating theory



A real-world example

- Oct. 2000, Xerox was in cash crunch and tapped a \$7 billion bank credit line after failed to issue commercial paper.
- Dec. 2, 2000, Xerox's Bond was downgraded to Junk rating Ba1
- One day in Oct. 2000, I was congratulated by my boss at one of largest banks of US
 - ✓ I was able to predict the risk of Xerox 5 Months prior to the October event using a combination of analytical tools including a PIT PD calculator

A real-world example



Credit Rating theory

- I was able to revise EDF methodology and produce stable ratings that is consistent with Bank's TTC requirements
 - ✓ My research was accepted by and presented to multiple Finance conferences including the World Finance Conference (Norway 2021) and the First Annual Volatility Institute Conference at New York University (2019)
 - ✓ My method is called Time-Consistent (TiC) approach that can unify all credit ratings (not just KMV rating)
 - My course: "Unified Credit Ratings" applies my method to all ratings
 - ✓ In the current course, we will learn both the KMV and the TiC ratings for public companies

A Universal TiC Rating Model

- Current Risk Rating Participants
 - Agency Ratings
 - ✓ Moody's/S&P/Fitch/FICO
 - Bank's Internal Risk Ratings
 - ✓ around 20 grades
 - Academic Researchers
 - ✓ Structural Models, other models
 - Regulators and Regulations
 - ✓ Especially after financial crisis
 - Credit Risk Analysts and Managers
 - ✓ Practitioners

A Universal TiC Rating Model

- Current Risk Rating Issues
 - Each use its own definitions, interpretations and methodologies, they are not consistent.
 - Difficult to completely translate/convert outcomes between them
 - ✓ PIT/TTC/Long/short/PD
 - We know there are differences among them, but we do not have realizable methods to measure their differences
 - Regulations are changing rapidly, we do not understand or incorporate regulatory impacts on credit ratings
 - There are many vague concepts that needs rigorous mathematical descriptions

A Universal TiC Rating Model

- An Example of the Issues
 - Moody's Historical B-Ratings and their default rates

Moody's B	2001	2002	2003	2004	2005	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016
Historical PD	9.8%	5.4%	2.8%	0.7%	1.3%	0.0%	2.1%	7.6%	0.5%	0.1%	0.5%	0.9%	0.3%	2.3%	1.6%

- After the financial crisis, several rating agencies had to pay large amounts of penalties.
- In 2012, US SEC issued a “Credit Rating Standardization Study” to US Congress

A Universal TiC Rating Model

- US SEC Report in 2012
 - It is desirable for a credit rating agency to have a standardized credit rating terminology...., However, ... may not be feasible given the number and uniqueness of rating scales and differences in credit rating methodologies used by credit rating agencies
- I set an objective for myself:
 - To establish a comprehensive credit rating theory and methodology that can address all the issues discussed early.
 - It was completed in 2021 after 10 years of efforts
- I named it as Time-Consistent (TiC) Credit Rating
- It can generated all credit ratings using a single approach and method
- It can be easily converted to and compared with TTC credit ratings by Banks and agencies

A Universal TiC Rating Model

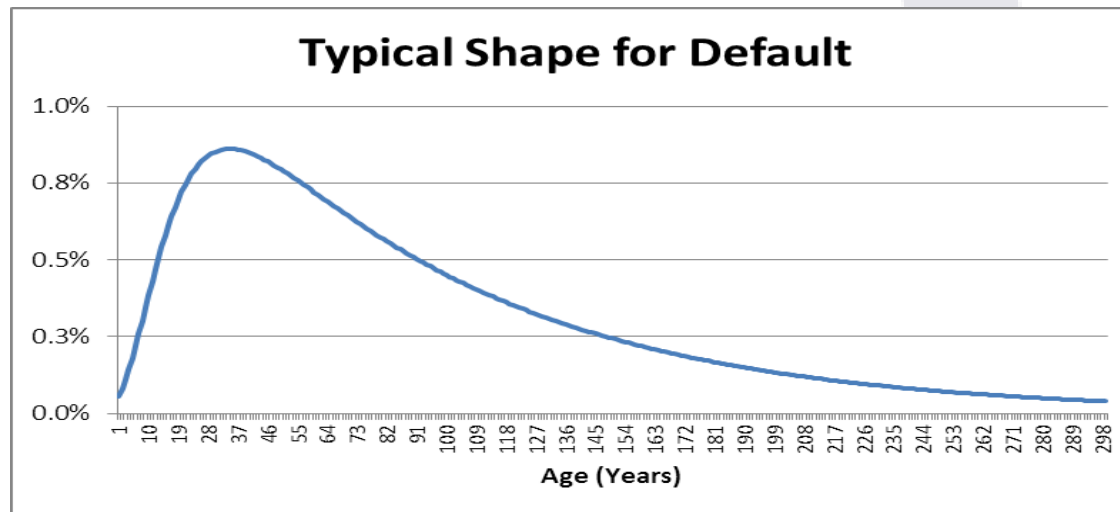
- Time-Consistent (TiC) Credit Rating was accepted and presented to
 - World Finance Conference 2021, Norway
 - Fifth Annual Volatility Institute Conference at New York University – Credit Risk in a New Era, November 22, 2019
 - 2018 IRFRC-APRIA Joint Conference at Nanyang Technological University of Singapore
 - 2015 International Conference on Financial Engineering & Innovation, ChengDu, China
 - Many academic institutions
 - ✓ Hong Kong University of Science and Technology
 - ✓ Academy of Mathematics & System Science, Chinese Academy of Sciences
 - ✓ University of Chinese Academy of Sciences

Behavior-Specific Approach & Two Driving Factors

- Two driving factors for Behavior-Specific Approach
 - One drives expected loss
 - The other drives unexpected loss
- Also drives credit behaviors
 - Level of Defaults
 - Deterioration of Credit
- Reflecting two main regulatory requirements
 - Allowance
 - Capital (this is addressed by none of existing rating approaches)
- Plus the interactions between the two drivers
- A common mistake made by many: Risk Rating is the same as Probability of Default (PD)

Factor One

- Every company will eventually default
 - With different life spans
 - ✓ Life Expectancy
 - ✓ Uncertainty around Life Expectancy
 - Let τ be the default time,
 - ✓ Its Expectation, $\mu = E[\tau]$ is the Expected Default Time. It is our first factor



Factor Two

- On the other hand, Credit quality can deteriorate or migrate
 - Like Depreciation or Corrosion in health
 - ✓ But how do we measure this Corrosion?
 - ✓ I created a Credit Corrosion Measure (CCM)
 - Let τ be the default time, our second factor is
$$CCM = E[\tau] \cdot E[1/\tau] - 1$$
 - Life Span $E[\tau]$ drives (Expected) PD
 - CCM drives credit deterioration

Time-Consistent (TiC) Credit Rating

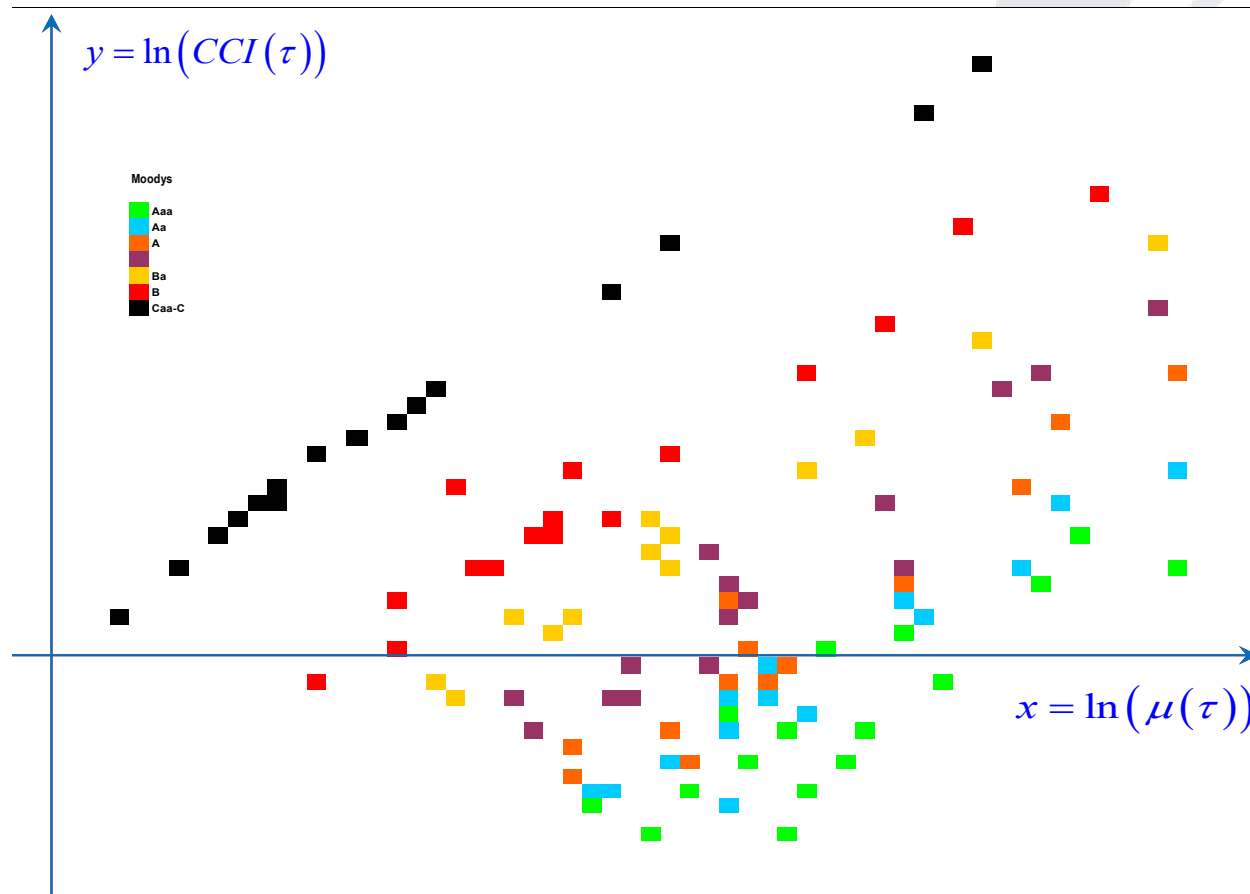
- Our Time-Consistent (TiC) Rating is then defined mathematically as the following

$$TiC \text{ Credit Rating} = \frac{CCM}{\mu^Q} = \frac{E[\tau] E\left[\frac{1}{\tau}\right] - 1}{E^Q[\tau]}$$

- Question One: does this rating solve all problems mentioned earlier?
 - ✓ Answer: Yes, and a lot more
- Question Two: Are the three main rating approaches can be represented by/regarded as TiC Ratings?
 - ✓ Answer: Yes, they are all special cases of TiC ratings.

A Universal TiC Rating Model

- Time-Consistent (TiC) Credit Rating applied to Agency Ratings



A Universal TiC Rating Model

- Time-Consistent (TiC) Credit Rating (x100) applied to Agency Ratings

S&P	AAA	RS	AA	RS	A	RS	BBB	RS	BB	RS	B	RS	CCC/C	RS
2005	0.00%	1.9	0.00%	2.2	0.00%	4.1	0.13%	7.7	0.4%	18	2.5%	56	13.1%	175
2006	0.00%	1.8	0.00%	3.2	0.00%	7.1	0.00%	10.4	0.4%	20	0.7%	42	17.5%	186
2007	0.00%	1.5	0.00%	2.9	0.00%	5.1	0.00%	8.9	0.4%	18	0.1%	41	19.4%	184
2009	0.00%	2.1	0.00%	3.4	0.19%	4.6	0.58%	8.3	0.9%	17	11.0%	52	57.3%	96
2010	0.00%	3.5	0.00%	3.6	0.00%	3.8	0.00%	8.1	0.0%	22	1.3%	67	25.2%	195
2011	0.00%	3.1	0.00%	3.2	0.00%	3.6	0.14%	5.7	0.0%	15	1.8%	61	18.3%	94
2012	0.00%	1.7	0.00%	2.6	0.00%	3.5	0.00%	7.1	0.0%	17	1.2%	46	33.7%	169
2013	0.00%	1.2	0.00%	2.7	0.00%	7.1	0.00%	11.8	0.0%	22	1.0%	55	33.3%	195
2014	0.00%	2.5	0.00%	2.6	0.00%	2.7	0.00%	6.4	0.0%	15	0.6%	43	33.3%	180
2015	0.00%	1.2	0.00%	2.4	0.00%	4.8	0.00%	8.4	0.2%	19	2.7%	39	36.7%	133
2016	0.00%	2.9	0.00%	3.0	0.00%	4.8	0.00%	11.2	0.6%	22	4.2%	51	46.0%	144
Avg	0.00%	2.7	0.03%	3.5	0.07%	5.2	0.23%	9.9	0.9%	22	4.4%	51	33.6%	155

A Universal TiC Rating Model

- Time-Consistent (TiC) Credit Rating applied to Banks Ratings

$$\ln(\text{CreditRating}) = 0.75 \cdot \Phi^{-1}(PD_1) \cdot \sqrt{-\frac{2}{3} \ln(\lambda)} + \frac{0.75}{3} \cdot \ln(\lambda) + \ln\left(\frac{1}{\lambda^{\frac{2}{3}}} - 1\right)$$

- By a mathematical formula
- PD is clearly different from the rating. The TiC rating reflects PD with different time periods

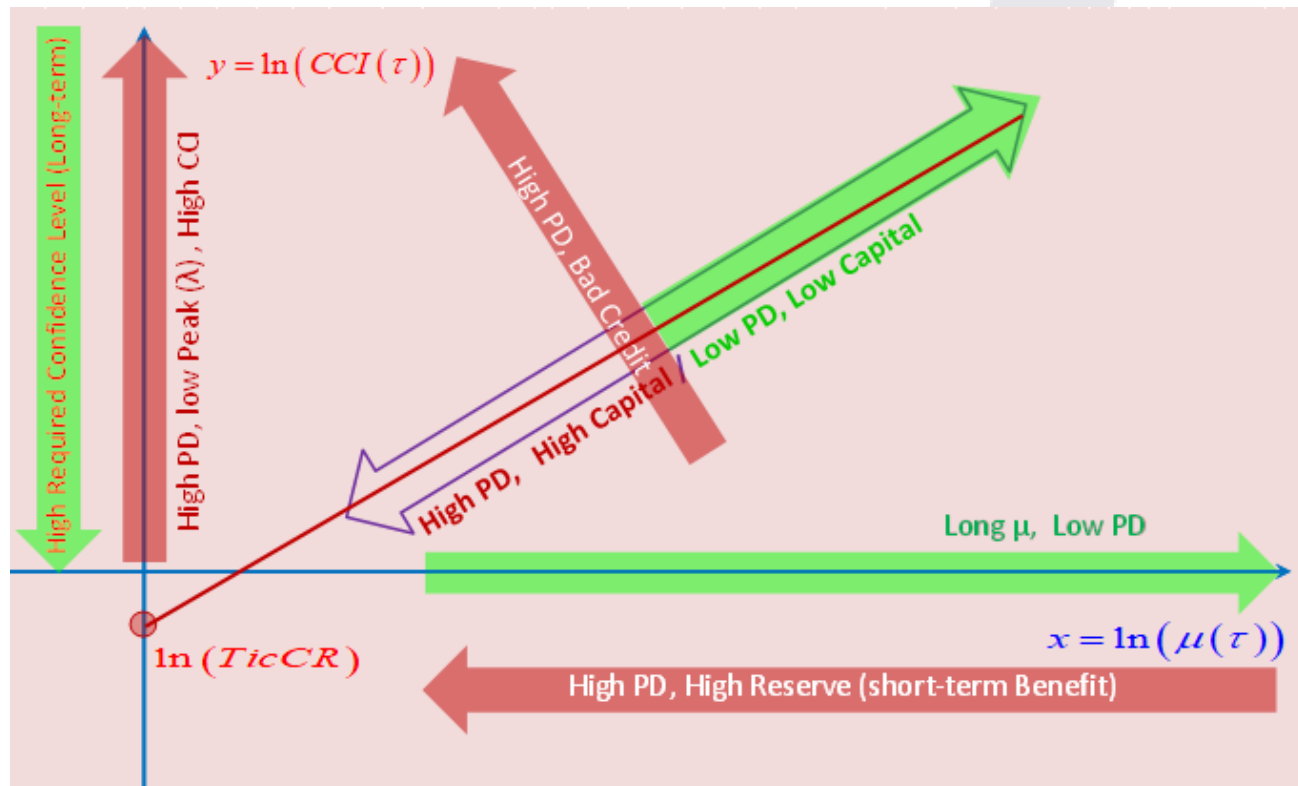
A Universal TiC Rating Model

- Time-Consistent (TiC) Credit Rating applied to Market-driven Structural Model (PIT rating)

$$CreditRating = \frac{CCM}{E^1[\tau]} = \frac{\sigma_A^2}{\ln^2\left(\frac{A_0}{D}\right)}$$

A Universal TiC Rating Model

- Rating Conversions between these ratings become simple and easy
- Even government policies can be quantified



KMV Rating Method

- It consists of the following components
 - Theory and Assumptions
 - Parameters
 - ✓ Observable Parameters
 - ✓ Non-observable parameters
 - Relationship and Equations
 - Inputs
 - ✓ Financial Inputs
 - Inputs adjustments
 - ✓ Market Inputs
 - Algorithms
 - Outputs

KMV Rating Method

- Theory and Assumptions
 - Merton: Equity is a Call Option on Asset with Debt being strike
 - Efficient Market Hypothesis: Non-arbitrage and Risk Neutral Pricing
 - Asset follows Geometric Brownian Motion: at time t , the asset value A_t is given by

$$A_t = A_0 \cdot e^{\eta_A \cdot t + \sigma_A \cdot W_t}$$

- W_t is a Brownian Motion, σ_A is the Asset volatility, η_A is the Asset Return, A_0 is the initial Asset

KMV Rating Method

- Parameters
 - A_t is the market value of the company Asset at time t.
Unobservable.
 - Asset volatility σ_A represents the business risk of the company.
Unobservable.
 - η_A is the Asset Return. **Unobservable**
 - Debt D is the company's debt, **Observable**
 - ✓ It has two components: short-term debt and long-term debt.
 - ✓ Short-term debt plays bigger role than long-term debt
 - r is risk-free interest rate, **Observable**

KMV Rating Method

- Relationship and Equations
 - Stock/Equity E is given by the following equation (t=time to Debt due)

$$E = A_0 \cdot \Phi \left(\frac{\ln \left(\frac{A_0}{D} \right) + \left(r + \frac{\sigma_A^2}{2} \right) t}{\sigma_A \sqrt{t}} \right) - D \cdot e^{-rt} \cdot \Phi \left(\frac{\ln \left(\frac{A_0}{D} \right) + \left(r - \frac{\sigma_A^2}{2} \right) t}{\sigma_A \sqrt{t}} \right)$$

- Two unobservable parameters A_0 and σ_A
 - ✓ One method is to use another equation so that two equations can solve two unknowns. But the solutions are less stable
 - ✓ Another method is to use estimated A's to calculate its standard deviation to estimate σ_A through an algorithm

KMV Rating Method

- Inputs
 - Financial Inputs
 - ✓ Debt: short-term debt + long-term debt
 - ✓ Inputs adjustments
 - Short-term Debt is more important, so we take
$$D = (\text{short-term debt}) + 0.5 (\text{Long-term debt})$$
 - Assuming D is due in 1 year
 - Market Inputs
 - ✓ Stock Price
 - Input Adjustment: add back dividends not reflected on the balance sheet to Stock price to reflect total Equity return
 - ✓ Total Number of Shares
 - ✓ Risk-free interest rate (1 year)

KMV Rating Method

- Algorithm
 - E-M Steps repeated
 - ✓ E-Step: Calculate Asset A assuming σ_A
 - ✓ M-Step: Calculate Standard Deviation of Asset A to estimate new σ_A
 - Return on Asset η_A can also be calculated simultaneously
- Outputs
 - The main output is Asset volatility σ_A
 - The other two outputs are Asset value A and Return on Asset η_A

KMV Rating Method

- KMV Rating Formulae
 - Distance-to-Default DD Score

$$DD = \frac{\ln\left(\frac{A_0}{D}\right) + \left(\eta_A - \frac{\sigma_A^2}{2}\right)}{\sigma_A}$$

- Expected-Default-Frequency PD

$$EDF = \Phi(-DD) = \Phi\left(-\frac{\ln\left(\frac{A_0}{D}\right) + \left(\eta_A - \frac{\sigma_A^2}{2}\right)}{\sigma_A}\right)$$

- η_A makes both unstable

TIC Rating Method

- Time-Consistent (TiC) Rating Formula

$$TiC = \frac{\sigma_A^2}{\ln^2 \left(\frac{A_0}{D} \right)}$$

Deriving KMV and TIC ratings

KMV Rating Generation

- Obtain Data
 - Select a public company
 - Download
 - ✓ Time-series of its Stock prices
 - ✓ Total Number of shares outstanding
 - ✓ Recent quarterly financials including Balance Sheet
 - may need financials for past several quarters
 - ✓ Recent 1 year risk-free rates for several quarters

KMV Rating Generation

- Data Processing

- For each downloaded quarter

- ✓ Use the adjusted closed price (adjusted for splits and dividends)

- ✓ Calculate Equity Value for each day as

$$\text{Equity} = (\text{Stock Price}) \times (\text{Total Number of Shares})$$

- ✓ Calculate Debt for each quarter as

$$D = (\text{Short-term Debt}) + 0.5 \times (\text{Long-term Debt})$$

- ✓ Each year has about 250 trading days, so each day is about 1/250 years

- We only count trading days (roughly, workdays)

- Discounted Value: Given a discount rate r , for a cash C at time T_1 , its discounted value at time T_0 is given by

$$\tau = \frac{\# \text{ of trading days between } T_0 \text{ and } T_1}{250} \quad \tau \approx \frac{T_1 - T_0}{365} \quad \text{Discounted Value of } C \text{ at } T_0 = \frac{C}{(1+r)^\tau}$$

KMV Rating Generation

- Processed Data
 - Equity series for trading **calendar dates** $(t_1, t_2, \dots, t_N)$

$$Equity_* = \{E_{t_1}, E_{t_2}, \dots, E_{t_N}\}$$

- E2D ratio Series: assume that for each trading day t_i , its prior quarter statement **date** is $t_i(Q)$ with Debt $D_{t_i(Q)}$ and risk-free rate $r_{t_i(Q)}$

✓ Use the prior financials (and the risk free-rate) closest to the trading

$$E2D_* = \left\{ \begin{array}{l} \text{date} \\ z_{t_1} = \frac{E_{t_1}}{D_{t_1(Q)} \cdot e^{-r_{t_1(Q)} \cdot \tau_{t_1}}}, z_{t_2} = \frac{E_{t_2}}{D_{t_2(Q)} \cdot e^{-r_{t_2(Q)} \cdot \tau_{t_2}}}, \dots, z_{t_N} = \frac{E_{t_N}}{D_{t_N(Q)} \cdot e^{-r_{t_N(Q)} \cdot \tau_{t_N}}} \end{array} \right\}$$

$$\tau_{t_i} \approx 1 - \frac{t_i - t_i(Q)}{365}$$

KMV Rating Generation

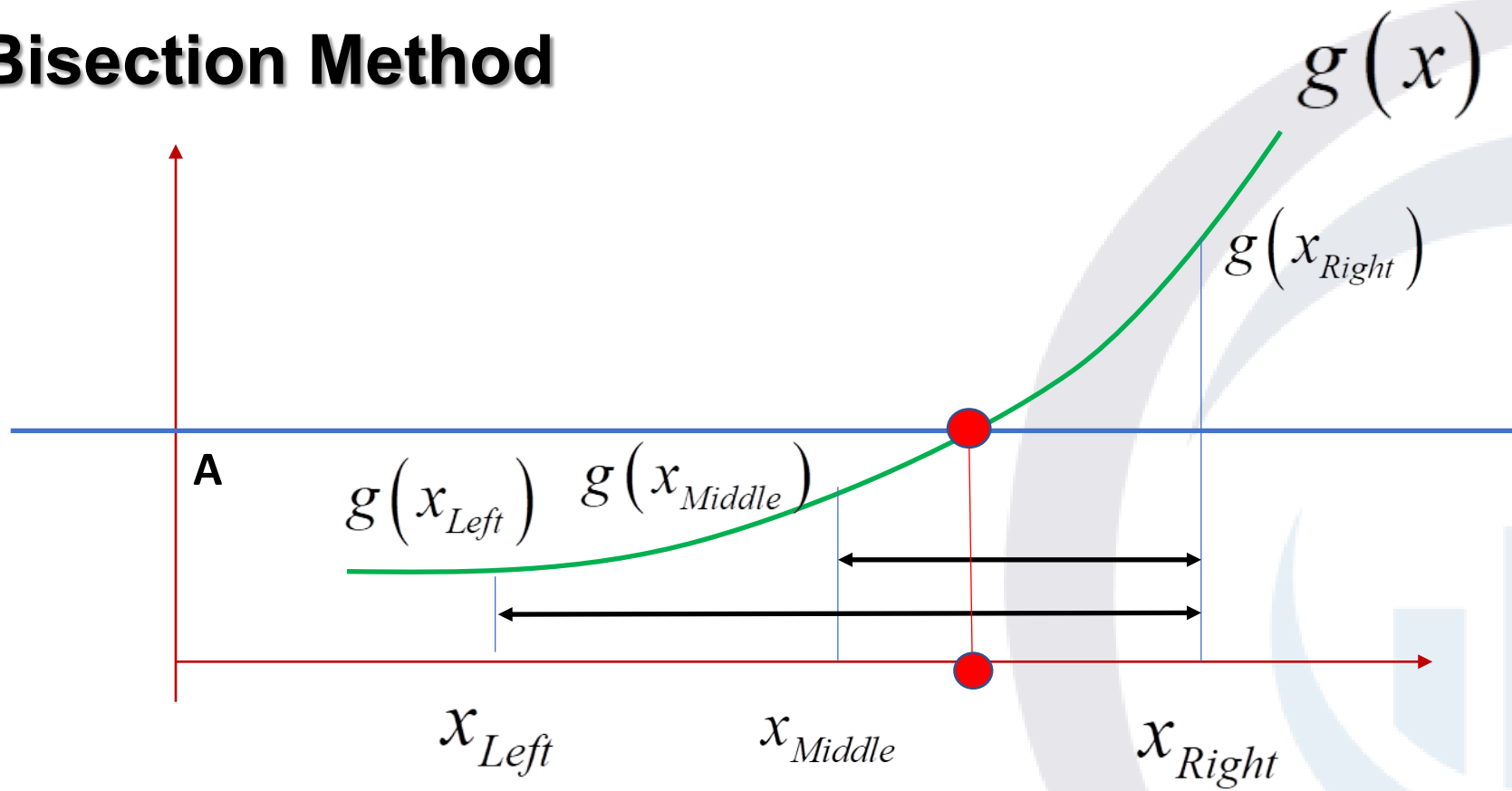
- A special function $g(x, \sigma, \tau)$

$$g(x, \sigma, \tau) = x \cdot \Phi\left(\frac{\ln(x) + \frac{\sigma^2}{2}\tau}{\sigma\sqrt{\tau}}\right) - \Phi\left(\frac{\ln(x) - \frac{\sigma^2}{2}\tau}{\sigma\sqrt{\tau}}\right)$$

- $g(x, \sigma, \tau)$ increases in x so we can easily find its inverse (using for example, Bisection method, see next page)

$$g^{-1}(x, \sigma, \tau)$$

Bisection Method



$$= \frac{x_{Left} + x_{Right}}{2}$$

is the new x_{Left}

KMV Rating Generation

- Expectation-Maximization (EM) Algorithm
 - **Initial Step**: define, for trading dates, initial values

$$\begin{cases}
 A_{t_i}^{(0)}, R_{t_i}^{(0)}, x_{t_i}^{(0)}, \sigma_A^{(0)}, R_A^{(0)} \\
 A_{t_i}^{(0)} = E_{t_i} \\
 R_{t_i}^{(0)} = \sqrt{250} \cdot \ln \frac{A_{t_{i+1}}^{(0)}}{A_{t_i}^{(0)}} \\
 x_{t_i}^{(0)} = \frac{A_{t_i}^{(0)}}{D_{t_i}(\mathcal{Q}) \cdot e^{-r_{t_i}(\mathcal{Q}) \cdot \tau_{t_i}}} \\
 \sigma_A^{(0)} = \begin{cases} \text{Standard Deviation} \\ \text{of } \{R_{t_i}^{(0)}\}_{1 \leq i \leq N} \end{cases} \\
 R_A^{(0)} = \frac{\sum_{i=1}^{N-1} R_{t_i}^{(0)}}{N-1} = \text{Average}
 \end{cases}$$

KMV Rating Generation

- Expectation-Maximization (EM) Algorithm
 - Take $m=1$
 - **E-Step**: estimate Asset To Debt Ratio $x_{t_i}^{(m)}$ and Asset $A_{t_i}^{(m)}$ using $\sigma_A^{(m-1)}$ by solving the inverse of the g function

$$g\left(x_{t_i}^{(m)}, \sigma_A^{(m-1)}, \tau_{t_i}\right) = z_{t_i}$$

$$z_{t_i} = \frac{E_{t_i}}{D_{t_i(Q)} \cdot e^{-r_{t_i(Q)} \cdot \tau_{t_i}}}$$

$$\tau_{t_i} \approx 1 - \frac{t_i - t_i(Q)}{365}$$

$$A_{t_i}^{(m)} = x_{t_i}^{(m)} \cdot D_{t_i(Q)} \cdot e^{-r_{t_i(Q)} \cdot \tau_{t_i}}$$

KMV Rating Generation

- Expectation-Maximization (EM) Algorithm
 - **M-Step** update Asset Volatility $\sigma_A^{(m)}$ through the following

$$R_{t_i}^{(m)} = \sqrt{250} \cdot \ln \frac{A_{t_{i+1}}^{(m)}}{A_{t_i}^{(m)}}$$

$$\sigma_A^{(m)} = \left\{ \begin{array}{l} \text{Standard Deviation} \\ \text{of } \left\{ R_{t_i}^{(m)} \right\}_{1 \leq i \leq N} \end{array} \right.$$

$$R_A^{(m)} = \frac{\sum_{i=1}^{N-1} R_{t_i}^{(m)}}{N-1}$$

- Set $m = m+1$ and repeat **E-M** steps

KMV Rating Generation

- We will have

$$\sigma_A^{(m)} \rightarrow \sigma_A$$

$$A_{t_i}^{(m)} \rightarrow A_{t_i}$$

$$R_A^{(m)} \rightarrow R_A = \eta_A - \frac{1}{2} \sigma_A^2$$

KMV and TIC Rating Generation

- Consider the most recent quarter $t_N(Q)$ and the most recent trading calendar day t_N . Let $A = A_{t_N}$, $D = D_{t_N}(Q)$ then

$$TiC = \frac{\sigma_A^2}{\ln^2 \frac{A}{D}}$$

$$DD = \frac{\ln \frac{A}{D} + \left(\eta_A - \frac{1}{2} \sigma_A^2 \right) \tau^*}{\sigma_A \sqrt{\tau^*}} \approx \frac{\ln \frac{A}{D} + \left(\eta_A - \frac{1}{2} \sigma_A^2 \right)}{\sigma_A}$$

$$\tau^* \approx 1 - \frac{t_N - t_N(Q)}{365}$$

$$EDF = \Phi(-DD) = \Phi \left(\frac{\ln \frac{A}{D} + \left(\eta_A - \frac{1}{2} \sigma_A^2 \right) \tau^*}{\sigma_A \sqrt{\tau^*}} \right)$$

$$\approx \Phi \left(\frac{\ln \frac{A}{D} + \left(\eta_A - \frac{1}{2} \sigma_A^2 \right)}{\sigma_A} \right)$$

Credit rating examples

Credit rating examples

- We use Microsoft (MSFT) as an example
 - Balance Sheet Information as of Dec. 31, 2022

Microsoft (MFST)	
Date	12/31/2022
Stock Price	\$239.82
Shares Outstanding	7,440,000,000
Current Liabilities	\$81,718,000,000
Long Term Debt	\$44,119,000,000
Other Non Current Liabilities	\$16,479,000,000
Other Current Liabilities	\$12,802,000,000
Short-term Rate	4.5%
D =	\$124,819,000,000

Credit rating examples

- The Algorithms is very fast and starts to converge almost immediately)

Run	Asset Volatility Outcome
[1]	30.23%
[2]	28.42%
[3]	28.42%
[4]	28.42%
[5]	28.42%

- We derive MSFT's Market Value of Asset

	Market Value of Asset	Market Value of Equity
3/24/2023	\$2,208,019,746	\$2,087,440,800
3/27/2023	\$2,176,867,852	\$2,056,267,200
3/28/2023	\$2,168,333,562	\$2,047,711,200
3/29/2023	\$2,207,638,476	\$2,086,994,400
3/30/2023	\$2,233,997,794	\$2,113,332,000
3/31/2023	\$2,265,639,515	\$2,144,952,000

Credit rating examples

○ DD and EDF (PIT PD) KMV Ratings

Market Value of Asset		DD	PIT PD
3/24/2023	\$2,208,019,746	10.41	0.00%
3/27/2023	\$2,176,867,852	10.36	0.00%
3/28/2023	\$2,168,333,562	10.35	0.00%
3/29/2023	\$2,207,638,476	10.41	0.00%
3/30/2023	\$2,233,997,794	10.46	0.00%
3/31/2023	\$2,265,639,515	10.51	0.00%

○ TIC Often uses TICx100 as score/rating

MSFT	3/31/2023
Asset Volatility	28.64%
Asset	\$2,265,617,704,382
Equity	\$2,144,951,910,720
Asset Growth Rate	12.85%
Distance-To-Default (DD)	10.426
PIT PD	0.0000%
TiC Rating	0.08531
TTC PD	0.0001%

Credit rating examples

- TIC can be converted to S&P TTC PD